

SPHERE: Mitigating the Loss of Spectral Plasticity in Mixture-of-Experts for Deep Reinforcement Learning

Anonymous Authors¹

Abstract

In DRL, an agent is trained from a stream of experience. In a continuous learning setting, such agents can suffer from *plasticity loss*: their ability to learn new skills from new experience diminishes over training. Recently, Mixture-of-Experts (MoE) networks are reported to enable scaling laws and facilitate the learning of diverse skills. However, in continual reinforcement learning settings, their performance can degenerate as learning proceeds, indicating a loss of plasticity. To address this, building on Neural Tangent Kernel (NTK) theory, we formalize the plasticity loss in MoE policies as a loss of *spectral plasticity*. We then derive a tractable proxy for spectral plasticity, one expressible in terms of individual expert feature matrices. Leveraging this proxy, we introduce *SPHERE*, a practical Parseval penalty tailored for MoE-based policies that alleviates the loss of spectral plasticity. On MetaWorld and HumanoidBench, SPHERE improves average success under continual RL by 133% and 50% over an unregularized MoE baseline, while maintaining higher spectral plasticity throughout training.

1. Introduction

In neuroscience and cognitive science, *plasticity loss* refers to a gradual decline in a learner’s ability to incorporate new experience (Livingston, 1966; Mateos-Aparicio & Rodríguez-Moreno, 2019). Recent work has shown that deep reinforcement learning (DRL) agents can exhibit a similar effect. As training proceeds, it becomes harder for the policy to improve from new experience (Kumar et al., 2021; Nikishin et al., 2022; Lyle et al., 2022; 2023; Abbas et al., 2023; Dohare et al., 2024; Sutton, 2018). This plasticity loss limits the ability of DRL agents to perform continual

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

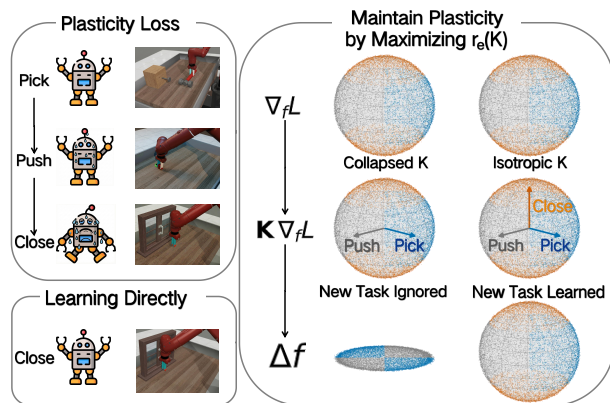


Figure 1. **Plasticity loss as loss of spectral plasticity.** **Left:** Continual RL can fail on later tasks despite isolated learnability. **Right:** By Eq. (7), $\Delta f = -\eta \mathbf{K} \nabla_f L$; low effective rank of the empirical Neural Tangent Kernel (eNTK) restricts updates to a few directions (collapsed spectrum), while high eNTK effective rank enables diverse directions (isotropic spectrum).

learning in continuous learning settings.

Meanwhile, Mixture-of-Experts (MoE) architectures provide a standard way to scale policy capacity in DRL and have been deployed in multi-task control for manipulators (Huang et al., 2025b), humanoids (Wang et al., 2025), and quadrupeds (Huang et al., 2025a), as well as in large-scale online RL systems (Obando-Ceron et al., 2024). Yet empirical studies report that, in continual learning settings, MoE performance can degrade as training proceeds (Willi et al., 2024), suggesting that MoE itself can suffer severe plasticity loss.

Plasticity loss in DRL has been increasingly studied. However, its manifestation in MoE policies remains poorly characterized. Motivated by this gap, we focus on mitigating plasticity loss in MoE policies. Addressing this problem is important because MoE policies are increasingly used in modern RL, yet plasticity loss can cause systematic failures on later tasks even when those tasks are learnable in isolation.

In this paper, we formalize plasticity loss through *spectral plasticity*: the diversity of functional update directions available to the policy, quantified by the spectral-entropy ef-

fective rank of the empirical Neural Tangent Kernel (eNTK). Under this lens, plasticity loss corresponds to a decline in this effective rank, which in turn confines functional updates to fewer directions. Since explicitly forming the eNTK matrix is intractable, we derive its layerwise block decomposition. This decomposition leads to a tractable effective-rank proxy controlled by a *weighted expert feature matrix* available from standard forward passes, yielding a principled, practical regularization target.

Guided by this view, we introduce *SPHERE*, Spectral Plasticity via Hyperspherical Expert REGularization, which applies a practical Parseval penalty to this weighted feature matrix. On MetaWorld and HumanoidBench, SPHERE improves average success by up to 133% and 50% over an unregularized MoE baseline, while maintaining higher spectral plasticity throughout training.

Concretely, we make the following contributions:

- We formalize plasticity loss as the loss of spectral plasticity and derive a tractable proxy building on NTK theory.
- We introduce SPHERE, a principled and practical regularizer to counter the loss of spectral plasticity.
- We evaluate SPHERE on MetaWorld and HumanoidBench, showing consistent gains over MoE baselines and prior plasticity methods, and sustained spectral plasticity throughout training.

2. Related Work

Plasticity Loss in DRL. Plasticity loss, also described as capacity loss (Lyle et al., 2022) and primacy bias (Nikishin et al., 2022), refers to a progressive decline in an agent’s ability to learn from new interaction data as training proceeds (Lyle et al., 2023; Nikishin et al., 2022). Empirically, it manifests as increased dormant neurons (Sokar et al., 2023), effective-rank and spectral collapse in representations (Chung et al., 2024), and related collapse in Hessian spectra (He et al., 2025). It is often attributed to the non-stationarity of RL training, where early experience can dominate and hinder learning from later samples (Igl et al., 2021; Kumar et al., 2021). In continual RL, these effects can lead to substantial performance degradation (Dohare et al., 2024). Recent work links plasticity loss to eNTK spectral collapse and proposes mitigation strategies in standard DRL architectures (Lyle et al., 2024; Tang et al., 2025). We build on this line of work but focus specifically on MoE policies.

MoE in Online DRL. MoE architectures have been adopted in online DRL as a means of scaling policy capacity via sparse expert routing (Obando-Ceron et al., 2024), and have been deployed in multi-task control for manipulators

(Huang et al., 2025b), humanoids (Wang et al., 2025), and quadrupeds (Huang et al., 2025a). Willi et al. (2024) further evaluate MoE in continual RL settings and report performance degradation over task sequences, indicating that MoE policies can suffer plasticity loss. Despite the broad use and study of MoE in RL, work on mitigating plasticity loss in MoE policies remains lacking.

3. Preliminaries

Markov Decision Processes. Consider a Markov Decision Process (MDP) defined by a tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \rho_0, T)$, with state space $\mathcal{S}$, action space $\mathcal{A}$, transition kernel $\mathcal{P} : \mathcal{S} \times \mathcal{A} \rightarrow \mathcal{P}(\mathcal{S})$, reward function $\mathcal{R} : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$, discount factor $\gamma \in [0, 1)$, initial-state distribution ρ_0 , and horizon T (Sutton, 2018). The agent interacts with the MDP via a policy $a_t \sim \pi(\cdot | s_t)$ that maps states to action distributions. The objective is to optimize the policy to maximize the expected discounted return $J(\pi) = \mathbb{E}_\pi[\sum_{t=0}^T \gamma^t r_t]$, where $s_0 \sim \rho_0$, $s_{t+1} \sim \mathcal{P}(\cdot | s_t, a_t)$ and $r_t = \mathcal{R}(s_t, a_t)$. The state-action value function is $q^\pi(s, a) = \mathbb{E}_\pi[\sum_{t=0}^T \gamma^t r_t | s_0 = s, a_0 = a]$ for $(s, a) \in \mathcal{S} \times \mathcal{A}$.

Top- K Mixture-of-Experts Policies. In a dense MoE (Shazeer et al., 2017), a set of experts $\{m_e(\cdot; \theta_e)\}_{e=1}^E$ is combined by a gate that produces mixture weights. Given an input x , each expert produces an output $m_e(x; \theta_e)$, and the gate computes logits $s(x; \psi) \in \mathbb{R}^E$ and normalized weights $h(x; \psi) \in \mathbb{R}^E$ (e.g., via softmax). The MoE output is the weighted sum

$$y(x; \Theta) = \sum_{e=1}^E m_e(x; \theta_e) h_e(x; \psi), \quad (1)$$

where $\Theta = (\psi, \{\theta_e\}_{e=1}^E)$ denotes all learnable parameters.

Top- K MoE introduces sparsity by routing each input to only a subset of experts (Shazeer et al., 2017). Given gate logits $s(x; \psi) \in \mathbb{R}^E$, define the selected expert index set $\mathcal{S}(x) \subseteq \{1, \dots, E\}$ as $\mathcal{S}(x) = \arg \text{top}_K(s(x; \psi))$, i.e., the indices of the K largest components of $s(x; \psi)$, with $|\mathcal{S}(x)| = K$. The corresponding subset softmax is

$$h_e^{(K)}(x) = \begin{cases} \frac{\exp(s_e(x))}{\sum_{j \in \mathcal{S}(x)} \exp(s_j(x))}, & \text{if } e \in \mathcal{S}(x), \\ 0, & \text{otherwise,} \end{cases} \quad (2)$$

and the MoE output is

$$y(x; \Theta) = \sum_{e \in \mathcal{S}(x)} m_e(x; \theta_e) h_e^{(K)}(x; \psi), \quad (3)$$

In this paper, we also consider DS-MoE (Dai et al., 2024), which splits experts more finely and includes always-active shared experts.

Empirical Neural Tangent Kernel. Consider a parameterized model $f(x; \theta) \in \mathbb{R}$ with parameters $\theta \in \mathbb{R}^P$. Given inputs $X = \{x_n\}_{n=1}^N$, let $f(X; \theta) \in \mathbb{R}^N$ stack the outputs on X , and let $\mathbf{J} \in \mathbb{R}^{N \times P}$ denote its Jacobian with respect to θ , whose n -th row is $(\nabla_{\theta} f(x_n; \theta))^{\top}$. Then the empirical neural tangent kernel (eNTK) matrix (Lyle et al., 2024) is

$$\mathbf{K} = \mathbf{J}\mathbf{J}^{\top} \in \mathbb{R}^{N \times N}, \quad (4)$$

which captures gradient correlations between samples. Here $\mathbf{K}$ is symmetric positive semidefinite (SPSD): $(\mathbf{J}\mathbf{J}^{\top})^{\top} = \mathbf{J}\mathbf{J}^{\top}$, and for any $v \in \mathbb{R}^N$, $v^{\top} \mathbf{K} v = v^{\top} \mathbf{J}\mathbf{J}^{\top} v = \|\mathbf{J}^{\top} v\|_2^2 \geq 0$. The eNTK also governs gradient descent dynamics in output space. Consider the loss $L(\theta) = \sum_{n=1}^N \ell(f(x_n; \theta))$ on the minibatch X , and denote $f_i(X) = f(X; \theta_i) \in \mathbb{R}^N$. Then discrete-time gradient descent induces the functional iteration (Lee et al., 2019)

$$f_{i+1}(X) = f_i(X) - \eta \mathbf{K}_i \nabla_{f_i(X)} L, \quad (5)$$

where $\mathbf{K}_i$ denotes the eNTK matrix on X at iteration i .

Continual RL and Plasticity Loss. In continual RL, an agent encounters a sequence of tasks $\mathcal{T}_1, \mathcal{T}_2, \dots, \mathcal{T}_k$, each modeled as an MDP. We assume these MDPs share the same state and action spaces but may differ in transition dynamics and reward functions. Recent work in DRL links plasticity loss to spectral collapse in the eNTK spectrum. Lyle et al. (2024) and Tang et al. (2025) demonstrate that the spectrum of the eNTK of dense MLP networks collapses to a low-rank structure as plasticity degrades.

Spectral-Entropy Effective Rank. Since rank is a discrete quantity, it is not amenable to gradient-based optimization and is highly sensitive to small perturbations. We therefore use the spectral-entropy effective rank (Roy & Vetterli, 2007) as a smooth proxy. For a matrix M with singular values $\{\sigma_i(M)\}$ and normalized weights $p_i = \sigma_i(M) / \sum_j \sigma_j(M)$, this spectral-entropy effective rank is

$$r_e(M) = \exp\left(-\sum_{\{i: \sigma_i(M) > 0\}} p_i \log p_i\right), \quad (6)$$

where $r_e(M) \leq \text{rank}(M) \leq \min\{m, n\}$. The first inequality holds with equality if and only if the nonzero singular values of M are equal; the second holds with equality if and only if M is full rank.

4. Method

In this section, we introduce *spectral plasticity* and use it to characterize plasticity loss in Top- K MoE policies, deriving a tractable surrogate objective. We first formalize spectral plasticity and relate its decay to plasticity loss in Section 4.1. To obtain a tractable objective without explicitly constructing the eNTK matrix $\mathbf{K}$, we analyze spectral plasticity in

parameter space and decompose the resulting matrix structure into smaller matrices. This yields a tractable surrogate objective, which we optimize to preserve spectral plasticity and motivates the SPHERE penalty in Section 4.2.

4.1. Formulating Spectral Plasticity

A spectral interpretation of plasticity loss. We first relate the functional change induced by gradient descent to the eNTK matrix $\mathbf{K}$. For brevity, we drop the iteration index, let $\Delta f(X) = f_{i+1}(X) - f_i(X)$ denote the change in outputs across two consecutive gradient steps, and write $\nabla_f L = \nabla_{f(X)} L$. Then Eq. (5) gives

$$\Delta f = -\eta \mathbf{K} \nabla_f L, \quad (7)$$

Since $\mathbf{K}$ is SPSPD, it admits the eigendecomposition $\mathbf{K} = \sum_{j=1}^N \lambda_j u_j u_j^{\top}$, where $\{u_j\}_{j=1}^N$ are orthonormal eigenvectors and $\lambda_j \geq 0$ are the corresponding eigenvalues. Substituting into Eq. (7) yields

$$\Delta f = -\eta \sum_{j=1}^N \lambda_j \langle u_j, \nabla_f L \rangle u_j, \quad (8)$$

where $\langle u_j, \nabla_f L \rangle$ is the projection of the loss gradient onto direction u_j . $\{u_j\}$ are the update directions and $\{\lambda_j\}$ are their gains. If $\lambda_j = 0$, then the component of $\nabla_f L$ along u_j is completely filtered out and cannot change the outputs, indicating a loss of learning capacity in that direction. More generally, when the spectrum is highly non-uniform, $\mathbf{K}$ biases updates toward a small set of directions, implicitly encoding a prior over gradient structure and attenuating components that do not align with the dominant eigenspace. Accordingly, high plasticity corresponds to an isotropic spectrum, with eigenvalues as equal as possible.

Definition 4.1 (Spectral Plasticity). Given inputs $X = \{x_i\}_{i=1}^N$, let $\mathbf{K}(X; \theta)$ denote the eNTK matrix (Eq. (4)). We define spectral plasticity as $\text{SP}(X; \theta) \triangleq r_e(\mathbf{K}(X; \theta))$.

Effective-Rank Equivalence Between $\mathbf{K}$ and G^{GN} . Directly working with the sample-space eNTK $\mathbf{K} = \mathbf{J}\mathbf{J}^{\top}$ is expensive to form and analyze (e.g., $O(N^2 P)$ to form $\mathbf{J}\mathbf{J}^{\top}$ given $\mathbf{J} \in \mathbb{R}^{N \times P}$), and it obscures the gate/expert and layerwise parameter partition needed to derive a tractable layerwise decomposition. We therefore work in parameter space via the Gauss–Newton matrix $G^{\text{GN}} = (1/N) \mathbf{J}^{\top} \mathbf{J}$ (Botev et al., 2017; Zhao et al., 2024). Since $\mathbf{K} = \mathbf{J}\mathbf{J}^{\top}$ and $\mathbf{J}^{\top} \mathbf{J}$ share the same nonzero eigenvalues (Horn & Johnson, 2012; Golub & Van Loan, 2013) and r_e is invariant under positive scalar rescaling (Roy & Vetterli, 2007), we have

$$r_e(\mathbf{K}) = r_e(\mathbf{J}^{\top} \mathbf{J}) = r_e((1/N) \mathbf{J}^{\top} \mathbf{J}) = r_e(G^{\text{GN}}). \quad (9)$$

Gate–Expert Block Decomposition of G^{GN} . Using Eq. (9), we analyze $r_e(\mathbf{K})$ through the Gauss–Newton matrix G^{GN} ;

directly maximizing $r_e(G^{\text{GN}})$ remains intractable, so we decompose G^{GN} into smaller gate and expert blocks and analyze them separately. For a Top- K MoE policy, we partition parameters into the gate and experts by writing $\theta = (\psi, \{\theta_e\}_{e=1}^E)$, where ψ denotes the gate parameters and θ_e denotes the parameters of expert e . Let L^g and L^{exp} denote the numbers of gate and expert layers, respectively.

For the Top- K MoE it is convenient to partition the parameter Jacobian $\mathbf{J} \in \mathbb{R}^{N \times P}$ according to gate and expert parameters:

$$\mathbf{J} = [\mathbf{J}^g \mid \mathbf{J}^{\text{exp}}], \quad (10)$$

where $\mathbf{J}^g \in \mathbb{R}^{N \times P^g}$ contains the columns corresponding to gate parameters ψ and $\mathbf{J}^{\text{exp}} \in \mathbb{R}^{N \times P^{\text{exp}}}$ contains the columns corresponding to expert parameters $\{\theta_e\}_{e=1}^E$.

Within the gate, we further decompose

$$\mathbf{J}^g = [\mathbf{J}_1^g \cdots \mathbf{J}_{L^g}^g], \quad (11)$$

where $\mathbf{J}_\ell^g \in \mathbb{R}^{N \times p_\ell^g}$ stacks the vectorized gradients with respect to the parameters of gate layer ℓ . For the experts, we similarly write

$$\mathbf{J}^{\text{exp}} = [\mathbf{J}_1^{\text{exp}} \cdots \mathbf{J}_{L^{\text{exp}}}^{\text{exp}}], \quad \mathbf{J}_\ell^{\text{exp}} = [\mathbf{J}_{1,\ell}^{\text{exp}} \cdots \mathbf{J}_{E,\ell}^{\text{exp}}], \quad (12)$$

where $\mathbf{J}_{e,\ell}^{\text{exp}} \in \mathbb{R}^{N \times p_{e,\ell}^{\text{exp}}}$ stacks the vectorized gradients with respect to the parameters of expert e at layer ℓ .

Under the gate/expert partition above, the GN matrix $G^{\text{GN}} = (1/N) \mathbf{J}^\top \mathbf{J}$ has the block form

$$G^{\text{GN}} = \frac{1}{N} \begin{bmatrix} (\mathbf{J}^g)^\top \mathbf{J}^g & (\mathbf{J}^g)^\top \mathbf{J}^{\text{exp}} \\ (\mathbf{J}^{\text{exp}})^\top \mathbf{J}^g & (\mathbf{J}^{\text{exp}})^\top \mathbf{J}^{\text{exp}} \end{bmatrix} = \begin{bmatrix} \mathbf{G}^{\text{GN},g} & \mathbf{G}^{\text{GN},g,\text{exp}} \\ \mathbf{G}^{\text{GN},\text{exp},g} & \mathbf{G}^{\text{GN},\text{exp}} \end{bmatrix}, \quad (13)$$

Here $\mathbf{G}^{\text{GN},g}$ and $\mathbf{G}^{\text{GN},\text{exp}}$ denote the gate and expert blocks, and $\mathbf{G}^{\text{GN},g,\text{exp}}$ and $\mathbf{G}^{\text{GN},\text{exp},g}$ denote the cross blocks. $(\mathbf{J}^g)^\top \mathbf{J}^g$ is an $L^g \times L^g$ block matrix whose (ℓ, ℓ') block is $(\mathbf{J}_\ell^g)^\top \mathbf{J}_{\ell'}^g$, and $(\mathbf{J}^{\text{exp}})^\top \mathbf{J}^{\text{exp}}$ is an $L^{\text{exp}} \times L^{\text{exp}}$ block matrix whose (ℓ, ℓ') block is $(\mathbf{J}_\ell^{\text{exp}})^\top \mathbf{J}_{\ell'}^{\text{exp}}$. Since $\mathbf{J}_\ell^{\text{exp}} = [\mathbf{J}_{1,\ell}^{\text{exp}} \cdots \mathbf{J}_{E,\ell}^{\text{exp}}]$, each same-layer expert block $(\mathbf{J}_\ell^{\text{exp}})^\top \mathbf{J}_\ell^{\text{exp}}$ is itself an $E \times E$ block matrix whose (e, e') block is $(\mathbf{J}_{e,\ell}^{\text{exp}})^\top \mathbf{J}_{e',\ell}^{\text{exp}}$. We next derive the corresponding layerwise gate and expert blocks.

Layerwise Decomposition of $r_e(\mathbf{K})$. We now express the eNTK effective rank in terms of the layerwise gate and expert GN blocks, yielding a decomposition of $r_e(\mathbf{K})$ across layers.

For a general block matrix, the spectral-entropy effective rank does not admit a simple decomposition across blocks. We therefore adopt a standard block-diagonal approximation

(Botev et al., 2017). Under this approximation, the Gauss–Newton matrix decomposes as

$$G^{\text{GN}} \approx \left(\bigoplus_{\ell=1}^{L^g} \mathbf{G}_\ell^{\text{GN},g} \right) \oplus \left(\bigoplus_{\ell=1}^{L^{\text{exp}}} \mathbf{G}_\ell^{\text{GN},\text{exp}} \right), \quad (14)$$

where $\mathbf{G}_\ell^{\text{GN},g}$ and $\mathbf{G}_\ell^{\text{GN},\text{exp}}$ denote the gate- and expert-layer blocks and $\bigoplus$ denotes the block direct sum. Under this block-diagonal form, we derive an effective-rank identity for block-diagonal matrices and use it to decompose $r_e(\mathbf{K})$ into a more tractable layerwise form in the propositions below.

Proposition 4.1 (Layerwise mixture form of $r_e(\mathbf{K})$). *Under (14), define nuclear norms*

$$s_\ell^g = \|\mathbf{G}_\ell^{\text{GN},g}\|_* \quad \text{and} \quad s_\ell^{\text{exp}} = \|\mathbf{G}_\ell^{\text{GN},\text{exp}}\|_*, \quad (15)$$

with total mass $S = \sum_m s_m^g + \sum_m s_m^{\text{exp}} > 0$ and mixture weights

$$\alpha_\ell^g = \frac{s_\ell^g}{S}, \quad \text{and} \quad \alpha_\ell^{\text{exp}} = \frac{s_\ell^{\text{exp}}}{S}. \quad (16)$$

Then the spectral-entropy effective rank of the eNTK satisfies

$$r_e(\mathbf{K}) \approx \exp \left(H(\alpha) + \sum_{\{\ell: s_\ell^g > 0\}} \alpha_\ell^g \log r_e(\mathbf{G}_\ell^{\text{GN},g}) + \sum_{\{\ell: s_\ell^{\text{exp}} > 0\}} \alpha_\ell^{\text{exp}} \log r_e(\mathbf{G}_\ell^{\text{GN},\text{exp}}) \right), \quad (17)$$

where $H(\alpha) = - \sum_{\ell: s_\ell^g > 0} \alpha_\ell^g \log \alpha_\ell^g - \sum_{\ell: s_\ell^{\text{exp}} > 0} \alpha_\ell^{\text{exp}} \log \alpha_\ell^{\text{exp}}$ is the Shannon entropy of the combined gate and expert mixture weights.

Proof. By (9), $r_e(\mathbf{K}) = r_e(G^{\text{GN}})$. Under the block-diagonal approximation (14), G^{GN} is block diagonal with blocks $\{\mathbf{G}_\ell^{\text{GN},g}\}_{\ell=1}^{L^g}$ and $\{\mathbf{G}_\ell^{\text{GN},\text{exp}}\}_{\ell=1}^{L^{\text{exp}}}$. Applying Lemma E.2 to this block-diagonal surrogate with blocks $\{\mathbf{G}_\ell^{\text{GN},g}\}$ and $\{\mathbf{G}_\ell^{\text{GN},\text{exp}}\}$ yields the stated mixture expression for $r_e(G^{\text{GN}})$, and hence for $r_e(\mathbf{K})$ under the approximation. $\square$

Moreover, since $r_e(M) \leq \text{rank}(M) \leq P(M)$, the maximal attainable effective rank of the gate block is limited by its parameter count P^g , which in typical Top- K MoE policies is much smaller than the expert parameter count P^{exp} . Thus the gate contribution to the mixture in Proposition 4.1 is limited; in the sequel we treat the gate terms as fixed and focus on the expert blocks $\{\mathbf{G}_\ell^{\text{GN},\text{exp}}\}$.

Connecting $r_e(\mathbf{K})$ and Expert Features. Proposition 4.1 suggests focusing on the expert blocks $\{\mathbf{G}_\ell^{\text{GN},\text{exp}}\}$, but directly maximizing $r_e(\mathbf{G}_\ell^{\text{GN},\text{exp}})$ is still intractable. We therefore express each $\mathbf{G}_\ell^{\text{GN},\text{exp}}$ in terms of a concatenated

weighted expert feature factor, with weights given by the Top- K gate, and a pre-activation backprop-gradient factor; under a standard within-layer independence approximation, the next proposition gives a block Khatri–Rao factorization between these two factors.

Proposition 4.2 (Feature–gradient factorization of $\mathbf{G}_\ell^{\text{GN},\text{exp}}$). *Consider an expert layer ℓ of the Top- K MoE policy (3). Under a standard within-layer independence approximation, the expert-layer block admits the block Khatri–Rao factorization*

$$\mathbf{G}_\ell^{\text{GN},\text{exp}} \approx \mathbf{A}_{\ell-1}^{\text{exp}} * \mathbf{G}_\ell^{\text{exp}}, \quad (18)$$

where $*$ denotes the block Khatri–Rao product on corresponding expert blocks. Write $a_{e,\ell-1}^{\text{exp}}(x_i) \in \mathbb{R}^{d_{e,\ell-1}^{\text{exp}}}$ for the input to expert e on sample x_i , $g_{e,\ell}^{\text{exp}}(x_i) \in \mathbb{R}^{d_{e,\ell}^{\text{exp}}}$ for the corresponding pre-activation backprop vector, and $h_{i,e}^{(K)}$ for the Top- K gating weight from (2). We define $\mathbf{A}_{\ell-1}^{\text{exp}}$ using the concatenated weighted expert feature vector

$$a_{\ell-1}(x_i) = [h_{i,1}^{(K)} a_{1,\ell-1}^{\text{exp}}(x_i)^\top \mid \cdots \mid h_{i,E}^{(K)} a_{E,\ell-1}^{\text{exp}}(x_i)^\top]^\top, \quad (19)$$

stacking them into $\Phi_{\ell-1} \in \mathbb{R}^{N \times (\sum_{e=1}^E d_{e,\ell-1}^{\text{exp}})}$ and setting

$$\mathbf{A}_{\ell-1}^{\text{exp}} = (1/N) \Phi_{\ell-1}^\top \Phi_{\ell-1}.$$

Similarly, we define $\mathbf{G}_\ell^{\text{exp}}$ by concatenating expert backprop vectors $g_\ell(x_i) = [g_{1,\ell}^{\text{exp}}(x_i)^\top \mid \cdots \mid g_{E,\ell}^{\text{exp}}(x_i)^\top]^\top$, stacking them into $\Psi_\ell \in \mathbb{R}^{N \times (\sum_{e=1}^E d_{e,\ell}^{\text{exp}})}$, and setting

$$\mathbf{G}_\ell^{\text{exp}} = (1/N) \Psi_\ell^\top \Psi_\ell.$$

Proof. See Appendix C.3. $\square$

Although Proposition 4.2 expresses each expert-layer block $\mathbf{G}_\ell^{\text{GN},\text{exp}}$ as a block Khatri–Rao product, the spectral-entropy effective rank $r_e(\mathbf{G}_\ell^{\text{GN},\text{exp}})$ remains difficult to analyze and optimize directly because the spectrum of a block Khatri–Rao product does not admit a simple characterization. We therefore introduce a Kronecker-product proxy $\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}} = \mathbf{A}_{\ell-1}^{\text{exp}} \otimes \mathbf{G}_\ell^{\text{exp}}$ and use its effective rank as a surrogate objective; the next proposition shows that maximizing the proxy effective rank is sufficient for maximizing $r_e(\mathbf{G}_\ell^{\text{GN},\text{exp}})$ under this model.

Proposition 4.3 (Optimal surrogate for $r_e(\mathbf{G}_\ell^{\text{GN},\text{exp}})$). *For each expert layer ℓ , let $\mathbf{G}_\ell^{\text{GN},\text{exp}} = \mathbf{A}_{\ell-1}^{\text{exp}} * \mathbf{G}_\ell^{\text{exp}}$ be the same-layer expert block in block Khatri–Rao form under this model, and let $\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}} = \mathbf{A}_{\ell-1}^{\text{exp}} \otimes \mathbf{G}_\ell^{\text{exp}}$ be its Kronecker proxy. Over all feature–gradient factors $(\mathbf{A}_{\ell-1}^{\text{exp}}, \mathbf{G}_\ell^{\text{exp}})$, every maximizer of the proxy effective rank $r_e(\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}})$ is also a maximizer of the expert-layer effective rank*

$r_e(\mathbf{G}_\ell^{\text{GN},\text{exp}})$; equivalently,

$$\arg\max_{\mathbf{A}_{\ell-1}^{\text{exp}}, \mathbf{G}_\ell^{\text{exp}}} r_e(\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}}) \subseteq \arg\max_{\mathbf{A}_{\ell-1}^{\text{exp}}, \mathbf{G}_\ell^{\text{exp}}} r_e(\mathbf{G}_\ell^{\text{GN},\text{exp}}). \quad (20)$$

Proof. See Appendix D. $\square$

The next proposition shows that, under the Kronecker proxy, the effective rank of each expert-layer block factorizes into a feature term and a backprop-gradient term. Consequently, treating the backprop-gradient term as fixed, maximizing the proxy reduces to maximizing the weighted expert feature effective rank, motivating our focus on $\mathbf{A}_{\ell-1}^{\text{exp}}$.

Proposition 4.4 (Global monotonicity via feature factors). *For each layer ℓ , the Kronecker proxy block is $\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}} = \mathbf{A}_{\ell-1}^{\text{exp}} \otimes \mathbf{G}_\ell^{\text{exp}}$. The effective rank factorizes as*

$$r_e(\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}}) = r_e(\mathbf{A}_{\ell-1}^{\text{exp}}) r_e(\mathbf{G}_\ell^{\text{exp}}), \quad (21)$$

so if the expert-gradient factors $\{\mathbf{G}_\ell^{\text{exp}}\}$ are fixed, maximizing $r_e(\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}})$ over $\mathbf{A}_{\ell-1}^{\text{exp}}$ is equivalent to maximizing $r_e(\mathbf{A}_{\ell-1}^{\text{exp}})$. Moreover, the global expert-block effective rank $r_e(\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}})$ is strictly increasing in each $r_e(\mathbf{A}_{\ell-1}^{\text{exp}})$.

Proof. See Appendix E. $\square$

Combined with Proposition 4.1, this motivates directly maximizing $r_e(\mathbf{A}_{\ell-1}^{\text{exp}})$ to increase the proxy effective rank and counter collapse of $r_e(\mathbf{K})$; Section 4.2 operationalizes this via the SPHERE regularizer on $\mathbf{A}_{\text{last}}^{\text{exp}}$.

4.2. Algorithm

We introduce the SPHERE regularizer and describe how it is integrated into PPO for Top- K MoE actors.

Objective. We regularize the weighted expert feature Gram at the last hidden expert layer of the MoE actor, denoted by $\mathbf{A}_{\text{last}}^{\text{exp}} \in \mathbb{R}^{m \times m}$, where $m = \sum_{e=1}^E d_{e,\text{last}}^{\text{exp}}$ is the concatenated feature dimension. We implement SPHERE with a scale-invariant Parseval-style Gram-isotropy penalty

$$\begin{aligned} \mathcal{L}_{\text{SPHERE}}(\mathbf{A}_{\text{last}}^{\text{exp}}) &= \min_{\alpha \in \mathbb{R}} \|\mathbf{A}_{\text{last}}^{\text{exp}} - \alpha \mathbf{I}_m\|_F^2 \\ &= \|\mathbf{A}_{\text{last}}^{\text{exp}}\|_F^2 - \frac{\text{Tr}(\mathbf{A}_{\text{last}}^{\text{exp}})^2}{m}. \end{aligned} \quad (22)$$

This quadratic is minimized at $\alpha^* = \text{Tr}(\mathbf{A}_{\text{last}}^{\text{exp}})/m$; see Appendix Lemma G.1.

Design. We now justify our regularizer by showing that pushing $\mathbf{A}_{\ell-1}^{\text{exp}}$ toward a scaled identity maximizes $r_e(\mathbf{A}_{\ell-1}^{\text{exp}})$; the following proposition formalizes this.

Proposition 4.5 (Maximizers of $r_e(\mathbf{A}_\ell^{\text{exp}})$). *For a PSD weighted expert feature Gram $\mathbf{A}_\ell^{\text{exp}} \in \mathbb{R}^{d_{\ell-1} \times d_{\ell-1}}$, the effective rank satisfies*

$$r_e(\mathbf{A}_\ell^{\text{exp}}) \leq d_{\ell-1}. \quad (23)$$

Equality holds iff $\mathbf{A}_\ell^{\text{exp}} = \alpha \mathbf{I}_{d_{\ell-1}}$ for some $\alpha > 0$.

Proof. See Appendix E. $\square$

Motivated by Proposition 4.5, we regularize $\mathbf{A}_{\text{last}}^{\text{exp}}$ toward a scaled identity using the SPHERE penalty. We act only on the actor to isolate effects on policy plasticity, and on this penultimate expert layer in particular: empirically, deeper layers tend to exhibit the lowest effective-rank embeddings (Huh et al., 2023), so maximizing the effective rank of this layer directly counteracts the dominant low-rank simplicity bias while leaving earlier layers free to adapt.

Total Loss and Integration. Let $\mathcal{L}_{\text{PPO}}$ denote the standard PPO objective. At each gradient update, we compute $\mathbf{A}_{\text{last}}^{\text{exp}}$ and add the SPHERE penalty $\mathcal{L}_{\text{SPHERE}}(\mathbf{A}_{\text{last}}^{\text{exp}})$ (Eq. (22)) to the actor.

$$\mathcal{L} = \mathcal{L}_{\text{PPO}} + \lambda^e \mathcal{L}_{\text{SPHERE}}(\mathbf{A}_{\text{last}}^{\text{exp}}). \quad (24)$$

λ^e is a regularization coefficient.

5. Experiments

We conduct experiments to characterize the loss of spectral plasticity in DRL and to evaluate SPHERE as a mitigation strategy. We first provide evidence of the phenomenon for both dense PPO and MoE policies, then evaluate SPHERE using task success and $r_e(\mathbf{K})$, study key design choices, and provide a qualitative analysis.

5.1. Experimental Setup

Training Protocols. We consider two training protocols: *RL*, where each task is trained independently from scratch, and *CRL*, where a single agent is trained sequentially through the list and training on each new task resumes from the final parameters of the previous task.

Tasks. We evaluate on tasks from two robotics benchmarks: HumanoidBench (Sferrazza et al., 2024) and MetaWorld (Yu et al., 2019). For HumanoidBench, we use the five H1 tasks (stand, walk, pole, slide, and run), since our CRL setting requires that all tasks share the same observation and action spaces. Meanwhile, tasks from MetaWorld (hammer, push-wall, faucet-close, push-back, stick-pull, handle-press-side, push, shelf-place, window-close, peg-unplug-side) evaluate fine-grained manipulation across diverse object interactions. For MetaWorld, we adopt the CW10 subset from Continual World (Wolczyk et al., 2021).

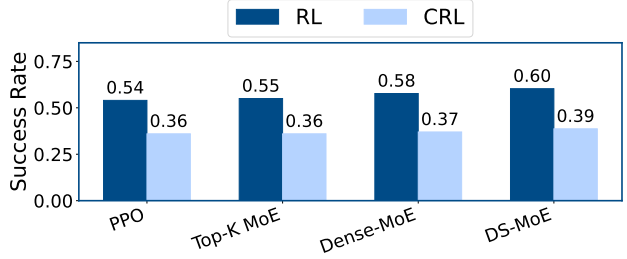


Figure 2. All methods show degraded performance under CRL relative to RL. Average final success rates on HumanoidBench under RL and CRL for PPO, Top- K MoE, Dense-MoE, and DS-MoE.

Training and Evaluation. Following standard benchmark protocols (Wolczyk et al., 2021; Sferrazza et al., 2024), we train for $T_{\text{MW}} = 10^6$ environment steps per MetaWorld task and $T_{\text{HB}} = 10^7$ steps per HumanoidBench task. We report the mean and standard deviation of final success rates over five random seeds. To quantify spectral plasticity, we additionally track $r_e(\mathbf{K})$; higher is better.¹

Baselines. We compare dense PPO, PPO(10x), which scales the actor network to $10\times$ the PPO actor parameter count, Top- K MoE actors, Dense-MoE actors, and DS-MoE (Dai et al., 2024). For Top- K MoE, we use $E = 10$ experts and route $K = 2$ experts per input. As alternative mitigation baselines, we apply C-CHAIN (Tang et al., 2025), which reduces churn to preserve plasticity, CBP (Dohare et al., 2024), which continuously reinitializes a subset of neurons, and PW (Chung et al., 2024), which regularizes weight matrices toward approximate orthogonality, to the same Top- K MoE actor used by SPHERE. All methods share the same PPO training pipeline.

5.2. Evidence of the Loss of Spectral Plasticity in DRL

We first show that the loss of spectral plasticity arises broadly in DRL across policy architectures. We compare RL with CRL for PPO, Top- K MoE, Dense-MoE, and DS-MoE, and report both final success rates and $r_e(\mathbf{K})$.

As shown in Figure 2, simply switching from RL to CRL reduces success for both dense PPO and MoE policies. Consistent with this, Figure 3 shows that MoE variants also exhibit rank collapse of $\mathbf{K}$, indicating the loss of spectral plasticity across MoE types. Together, these results show that the loss of spectral plasticity is a common phenomenon across policy architectures.

¹We evaluate $r_e(\mathbf{K})$ on a fixed held-out batch of states from rollouts of a pre-trained Stair policy (excluded from the CRL sequence) and reuse it across evaluation checkpoints and methods. This targets plasticity on *new* experience rather than on previously seen training states.

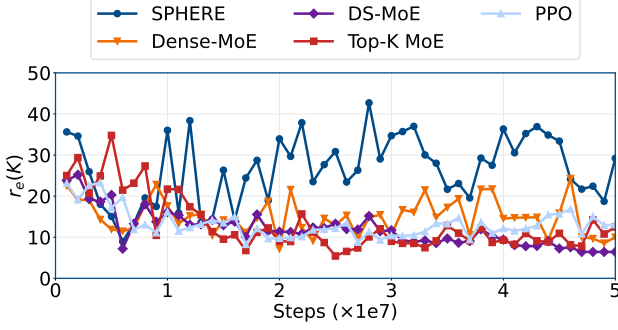


Figure 3. **SPHERE maintains higher $r_e(\mathbf{K})$ throughout training.** $r_e(\mathbf{K})$ during HumanoidBench CRL training for PPO, Top-K MoE, Dense-MoE, DS-MoE, and SPHERE. Across baselines, $r_e(\mathbf{K})$ drops substantially during training, indicating the loss of spectral plasticity.

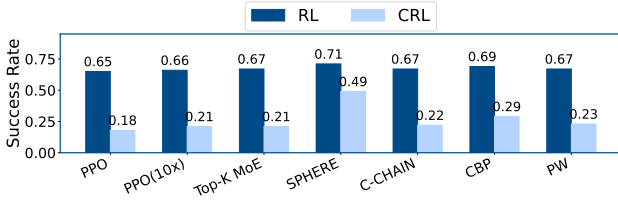


Figure 4. **Under CRL, SPHERE improves average success by 133% over Top-K MoE and reduces the RL–CRL gap by 52%.** Average final success rates on MetaWorld across methods under RL and CRL. SPHERE outperforms other mitigation baselines.

5.3. SPHERE Mitigates the Loss of Spectral Plasticity

We next evaluate whether SPHERE mitigates the loss of spectral plasticity, using both final success rates and $r_e(\mathbf{K})$.

Figures 4 and 5 summarize the average final success rates under RL and CRL on MetaWorld and HumanoidBench.

MetaWorld. On MetaWorld, SPHERE improves performance under both RL and CRL relative to the Top-K MoE baseline. Under CRL, SPHERE improves average success by +133% relative to Top-K MoE and substantially narrows the RL–CRL gap by 52%. The gain is larger under CRL, where sustained adaptation across tasks is required and preserving spectral plasticity helps maintain learning on later tasks.

HumanoidBench. On HumanoidBench, relative to Top-K MoE, SPHERE improves average success by 36% under RL and 50% under CRL. The large gain under RL shows that spectral plasticity can decay even within a single task. PPO continually updates from newly collected on-policy rollouts, so the effective training distribution drifts over time. HumanoidBench uses 10 $\times$ more environment steps per task than MetaWorld, which makes this within-task decay more pronounced and amplifies the benefit of SPHERE under RL.

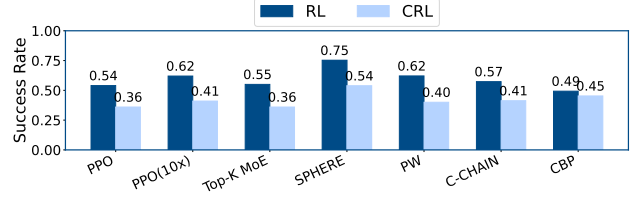


Figure 5. **Relative to Top-K MoE, SPHERE improves average success by 36% under RL and 50% under CRL.** Average final success rates on HumanoidBench across methods under RL and CRL. SPHERE outperforms other mitigation baselines.

Table 1. **The designs in Eq. (22) and Eq. (21) are both useful.** Ablation of SPHERE design choices on HumanoidBench under CRL. w/o SPHERE disables the SPHERE regularizer. Fixed α replaces the adaptive target scale in Eq. (22) with a constant. All hidden expert layers applies Eq. (22) to $\mathbf{A}_\ell^{\text{exp}}$ at all hidden expert layers ℓ . $\mathbf{G}_{\text{last}}^{\text{exp}}$ Regularization applies Eq. (22) to $\mathbf{G}_{\text{last}}^{\text{exp}}$ (Eq. (21)) rather than $\mathbf{A}_{\text{last}}^{\text{exp}}$. No variant has consistently good performance as SPHERE does.

Variant	Average success
w/o SPHERE	0.36 $\pm$ 0.08
w/ SPHERE	0.54 $\pm$ 0.12
Fixed $\alpha=1$	0.27 $\pm$ 0.09
Fixed $\alpha=2$	0.40 $\pm$ 0.18
Fixed $\alpha=5$	0.41 $\pm$ 0.09
All hidden expert layers	0.42 $\pm$ 0.07
$\mathbf{G}_{\text{last}}^{\text{exp}}$ Regularization	0.43 $\pm$ 0.09

5.4. Key Design Choices for SPHERE

We ablate key design choices of SPHERE on HumanoidBench under CRL with five seeds, as summarized in Table 1.

Adaptive Target Scale. SPHERE solves for the optimal scale α^* in closed form (Eq. (22)), avoiding the need to introduce and tune an additional scale hyperparameter for the regularizer and preventing the penalty magnitude from being mis-scaled relative to the PPO objective. Fixing α to a constant degrades performance relative to adaptive scaling, suggesting that treating the target scale as a fixed knob is not beneficial in this setting.

Layerwise Placement. SPHERE regularizes the last hidden expert layer of the actor. Applying the same penalty to all hidden expert layers underperforms regularizing only the last hidden expert layer, suggesting that regularizing earlier layers can overconstrain representation learning.

Gradient-Gram Regularization. By Eq. (21), the proxy effective rank factorizes into a feature term and a gradient term. As a complementary ablation, we apply the SPHERE penalty to $\mathbf{G}_{\text{last}}^{\text{exp}}$ while leaving $\mathbf{A}_{\text{last}}^{\text{exp}}$ unregularized. This improves over the baseline but remains below SPHERE, suggesting that the gains primarily come from shaping $\mathbf{A}_{\text{last}}^{\text{exp}}$. Since $\mathbf{G}_{\text{last}}^{\text{exp}}$ requires backpropagation to extract, we adopt feature-Gram regularization on $\mathbf{A}_{\text{last}}^{\text{exp}}$ as the main SPHERE

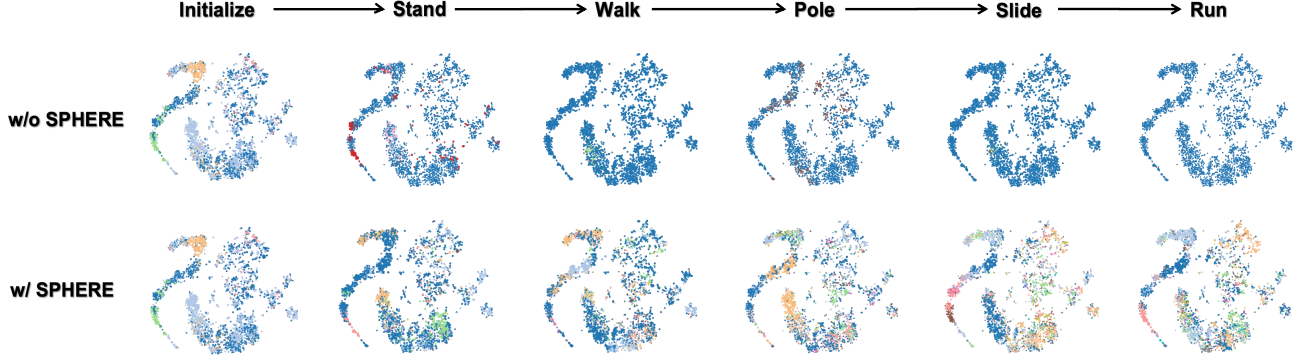


Figure 6. **SPHERE avoids collapse to a single component.** t-SNE embedding of held-out states. Colors indicate the dominant singular-direction index of the weighted expert feature matrix computed over the held-out batch, assigned per state by the largest squared projection. Columns show snapshots along the HumanoidBench CRL sequence. Top: w/o SPHERE. Bottom: w/ SPHERE.

instantiation.

5.5. Qualitative Analysis

Figure 6 gives a time-lapse view of effective-rank collapse in the weighted expert representations and the associated loss of spectral plasticity over the HumanoidBench CRL sequence. Using a fixed held-out state batch from Section 5.1, we visualize last-layer weighted expert features with t-SNE at initialization and after each task. Points are colored by the dominant singular-direction index of the weighted expert feature matrix computed over the held-out batch: for each state, we assign the singular direction with the largest squared projection of its feature vector.

In the baseline, colors quickly become nearly uniform, indicating that most states concentrate on a single dominant feature direction. By Eq. (21), this loss of feature diversity, with low $r_e(\mathbf{A}_{\ell-1}^{\text{exp}})$, directly shrinks $r_e(\tilde{\mathbf{G}}_{\ell}^{\text{GN,exp}})$ and, via Proposition 4.1, contributes to a lower $r_e(\mathbf{K})$. With SPHERE, multiple components remain active throughout the sequence, consistent with preserved spectral plasticity.

6. Conclusion

In this paper, we characterize the loss of spectral plasticity in DRL with mixture-of-experts policies and formalize it using the spectral-entropy effective rank of the actor empirical neural tangent kernel. We first establish that this loss arises consistently across different MoE architectures. We then propose SPHERE, a practical regularizer derived from our eNTK-based analysis, which mitigates the loss of spectral plasticity by encouraging isotropic weighted expert features. Through experiments on MetaWorld and HumanoidBench, we demonstrate that SPHERE consistently improves performance and maintains higher spectral plasticity.

Limitations. Our derivation relies on theoretical approximations. While we empirically validate these assumptions in

the appendix, removing these approximations is a direction for future theoretical work. We focus on MoE policies for continuous-control RL on MetaWorld and HumanoidBench; extending SPHERE to MoE architectures in large language models remains future work.

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

- Abbas, Z., Zhao, R., Modayil, J., White, A., and Machado, M. C. Loss of plasticity in continual deep reinforcement learning. In *Conference on Lifelong Learning Agents*, pp. 620–636. PMLR, 2023.
- Botev, A., Ritter, H., and Barber, D. Practical gauss-newton optimisation for deep learning. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *Proceedings of Machine Learning Research*. PMLR, 2017.
- Chung, W. et al. Parseval regularization for continual reinforcement learning. In *Advances in Neural Information Processing Systems*, 2024.
- Dai, D., Deng, C., Zhao, C., Xu, R. X., Gao, H., Chen, D., Li, J., Zeng, W., Yu, X., Wu, Y., Xie, Z., Li, Y. K., Huang, P., Luo, F., Ruan, C., Sui, Z., and Liang, W. Deepseekmoe: Towards ultimate expert specialization in mixture-of-experts language models, 2024.
- Dohare, S., Hernandez-Garcia, J. F., Lan, Q., Rahman, P., Mahmood, A. R., and Sutton, R. S. Loss of plasticity in deep continual learning. *Nature*, 632(8026):768–774, 2024.
- Golub, G. H. and Van Loan, C. F. *Matrix Computations*. Johns Hopkins University Press, 4 edition, 2013.
- Grosse, R. and Martens, J. A kronecker-factored approximate fisher matrix for convolution layers. In *Proceedings of the 33rd International Conference on Machine Learning*, 2016.
- He, N. et al. Spectral collapse drives loss of plasticity in deep continual learning. *arXiv preprint arXiv:2501.00001*, 2025.
- Horn, R. A. and Johnson, C. R. *Matrix Analysis*. Cambridge University Press, 2 edition, 2012.
- Huang, R. et al. Moe-loco: Mixture of experts for multitask locomotion. In *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025a.
- Huang, S. et al. Mentor: Mixture-of-experts network with task-oriented perturbation for visual reinforcement learning. In *International Conference on Machine Learning*, 2025b.
- Huh, M., Mobahi, H., Zhang, R., Cheung, B., Agrawal, P., and Isola, P. The low-rank simplicity bias in deep networks. *Transactions on Machine Learning Research*, 2023. Published in TMLR; OpenReview ID bCiN-WDmlY2.
- Igl, M. et al. Transient non-stationarity and generalisation in deep reinforcement learning. In *International Conference on Machine Learning*, 2021.
- Kumar, A., Agarwal, R., Ghosh, D., and Levine, S. Implicit under-parameterization inhibits data-efficient deep reinforcement learning. In *International Conference on Learning Representations*, 2021.
- Lee, J., Xiao, L., Schoenholz, S. S., Bahri, Y., Novak, R., Sohl-Dickstein, J., and Pennington, J. Wide neural networks of any depth evolve as linear models under gradient descent. *Advances in Neural Information Processing Systems*, 32, 2019.
- Liu, S. and Trenkler, G. Hadamard, khatri-rao, kronecker and other matrix products. *International Journal of Information and Systems Sciences*, 4(1):160–177, 2008.
- Livingston, R. B. Brain mechanisms in conditioning and learning. Technical report, Office of Naval Research, 1966.
- Lyle, C., Rowland, M., and Dabney, W. Understanding and preventing capacity loss in reinforcement learning. *arXiv preprint arXiv:2204.09560*, 2022.
- Lyle, C., Zheng, Z., Nikishin, E., Avila Pires, B., Pascanu, R., and Dabney, W. Understanding plasticity in neural networks. In *International Conference on Machine Learning*, pp. 23190–23211. PMLR, 2023.
- Lyle, C. et al. Disentangling the causes of plasticity loss in neural networks. *arXiv preprint arXiv:2401.00001*, 2024.
- Marshall, A. W., Olkin, I., and Arnold, B. C. *Inequalities: Theory of Majorization and Its Applications*. Springer, 2 edition, 2011.
- Martens, J. and Grosse, R. Optimizing neural networks with kronecker-factored approximate curvature. In *Proceedings of the 32nd International Conference on Machine Learning*, 2015.
- Mateos-Aparicio, P. and Rodríguez-Moreno, A. The impact of studying brain plasticity. *Frontiers in Cellular Neuroscience*, 13:66, 2019.
- Nikishin, E., Schwarzer, M., D’Oro, P., Bacon, P.-L., and Courville, A. The primacy bias in deep reinforcement learning. In *International Conference on Machine Learning*, volume 162, pp. 16828–16847. PMLR, 2022.
- Obando-Ceron, J. et al. Mixtures of experts unlock parameter scaling for deep rl. In *International Conference on Machine Learning*, 2024.

- Petersen, K. B. and Pedersen, M. S. The matrix cookbook, 2012. Version 20121115, Technical University of Denmark.
- Roy, O. and Vetterli, M. The effective rank: A measure of effective dimensionality. In *2007 15th European Signal Processing Conference*, pp. 606–610. IEEE, 2007.
- Sferrazza, C., Huang, D.-M., Lin, X., Lee, Y., and Abbeel, P. Humanoidbench: Simulated humanoid benchmark for whole-body locomotion and manipulation, 2024.
- Shazeer, N., Mirhoseini, A., Maziarz, P., Davis, A., Le, Q., Hinton, G., Dean, J., et al. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations*, 2017.
- Sokar, G., Agarwal, R., Castro, P. S., and Evci, U. The dormant neuron phenomenon in deep reinforcement learning. In *International Conference on Machine Learning*, pp. 32145–32168. PMLR, 2023.
- Sutton, R. S. Reinforcement learning: An introduction. A *Bradford Book*, 2018.
- Tang, H. et al. Mitigating plasticity loss in continual reinforcement learning by reducing churn. In *International Conference on Machine Learning*, 2025.
- Wang, D. et al. More: Mixture of residual experts for humanoid lifelike gaits learning on complex terrains. *arXiv preprint arXiv:2501.00000*, 2025.
- Willi, T. et al. Mixture of experts in a mixture of rl settings. *Reinforcement Learning Journal*, 2024.
- Wołczyk, M., Twardowski, B., Nowak, A., Rychlewska, B., and Trzciński, T. Continual world: A robotic benchmark for continual reinforcement learning, 2021.
- Yu, T., Quillen, D., He, Z., Julian, R., Hausman, K., Finn, C., and Levine, S. Meta-world: A benchmark and evaluation for multi-task and meta reinforcement learning. In *Proceedings of the Conference on Robot Learning*, 2019.
- Zhao, J., Singh, S. P., and Lucchi, A. Theoretical characterisation of the gauss newton conditioning in neural networks. In *Advances in Neural Information Processing Systems*, volume 37, pp. 114965–115000, 2024.

Algorithm 1 PPO with Actor MoE SPHERE regularization

- 1: Initialize actor π_θ with MoE layers and Top- K gating, and critic V_ϕ with standard dense layers.
 - 2: **for** iteration $j = 1$ to J **do**
 - 3: Collect trajectories with π_θ and compute advantages and PPO targets.
 - 4: **for** each PPO minibatch $\mathcal{B}$ of N samples **do**
 - 5: Run the MoE actor forward pass on $\mathcal{B}$ and compute $\mathbf{A}_{\text{last}}^{\text{exp}}$ at the last hidden expert layer.
 - 6: Compute $\mathcal{L}_{\text{PPO}}(\theta, \phi; \mathcal{B})$ and the SPHERE loss $\mathcal{L}_{\text{SPHERE}}(\mathbf{A}_{\text{last}}^{\text{exp}})$ via Eq. (22), then set the total loss $\mathcal{L}$ via Eq. (24).
 - 7: Update $\{\theta, \phi\}$ with a gradient step on $\mathcal{L}$.
 - 8: **end for**
 - 9: **end for**
-

A. Implementation Details

A.1. Adaptive Regularization Coefficient

In Eq. (24), λ^e trades off adherence to the SPHERE feature constraint against standard PPO optimization. We set it adaptively per minibatch. Unless stated, we use gradient-norm scaling,

$$\lambda^e = \rho \cdot \text{stopgrad} \left(\frac{\|\nabla_\theta \mathcal{L}_{\text{actor}}\|_2}{\|\nabla_\theta \mathcal{L}_{\text{SPHERE}}\|_2 + \varepsilon} \right),$$

where ρ is a tunable ratio and $\varepsilon > 0$ is a small constant.

A.2. Implementation and Training Algorithm

Algorithm 1 summarizes the resulting training procedure.

A.3. Implementation Details for Computing $r_e(\mathbf{K})$

Output definition. To measure spectral plasticity, we track the spectral-entropy effective rank $r_e(\mathbf{K})$ of the actor empirical NTK. For the diagonal-Gaussian actor used by PPO, we take $f_\theta(s) := \mu_\theta(s)$, where $\mu_\theta(s)$ is the mean vector of $\pi_\theta(a | s) = \mathcal{N}(\mu_\theta(s), \text{diag}(\sigma^2))$, and exclude the log-standard-deviation parameters. In our implementation, $\log \sigma$ is learned and state-independent.

SLQ computation of $r_e(\mathbf{K})$. We estimate $r_e(\mathbf{K})$ without explicitly constructing the sample-space Gram. Using Eq. (6), the spectral entropy admits the trace form

$$-\sum_k p_k \log p_k = \log \text{Tr}(\mathbf{K}) - \frac{\text{Tr}(\mathbf{K} \log \mathbf{K})}{\text{Tr}(\mathbf{K})},$$

Thus,

$$r_e(\mathbf{K}) = \exp \left(\log \text{Tr}(\mathbf{K}) - \frac{\text{Tr}(f(\mathbf{K}))}{\text{Tr}(\mathbf{K})} \right), \quad f(x) := x \log x. \quad (25)$$

Directly constructing $\mathbf{K}$ (or materializing $\mathbf{J} \in \mathbb{R}^{N \times P}$) is typically GPU-memory prohibitive for modern actor networks. We therefore estimate the traces in (25) with stochastic Lanczos quadrature (SLQ). For standard Gaussian probes $z \sim \mathcal{N}(0, I)$, Hutchinson’s identity gives $\text{Tr}(f(\mathbf{K})) = \mathbb{E}_z[z^\top f(\mathbf{K})z]$. Given a probe z , run M steps of the Lanczos algorithm on $\mathbf{K}$ with $q_1 = z/\|z\|_2$, using only matrix–vector products with $\mathbf{K}$, to obtain an orthonormal basis $Q_M = [q_1, \dots, q_M]$ and tridiagonal $T_M := Q_M^\top \mathbf{K} Q_M$. Gauss quadrature then yields the approximation

$$z^\top f(\mathbf{K})z \approx \|z\|_2^2 e_1^\top f(T_M) e_1, \quad (26)$$

where e_1 is the first standard basis vector. Averaging (26) over probes gives an estimator of $\text{Tr}(f(\mathbf{K}))$ (and similarly for $\text{Tr}(\mathbf{K})$ by taking $f(x) = x$), which we plug into (25). Finally, each Lanczos matrix–vector product can be computed as $\mathbf{K}v = \mathbf{J}(\mathbf{J}^\top v)$ via standard reverse-/forward-mode automatic differentiation, avoiding materialization of $\mathbf{K}$. For vector-valued outputs $f_\theta(x_i) \in \mathbb{R}^d$, we follow the SLQ estimator and reduce to the scalar-output form in Eq. (4) by defining a per-sample scalar projection $g_\theta(x_i) := \mathbf{1}_d^\top f_\theta(x_i)$. Let $\mathbf{J}$ in Eq. (4) be the Jacobian of g_θ over the batch, i.e., its i -th row is $(\nabla_\theta g_\theta(x_i))^\top$.

A.4. Interpreting the SPHERE Penalty

Let $\mathbf{A}_{\text{last}}^{\text{exp}} = (1/N)\Phi^\top \Phi$ denote the weighted expert feature Gram of the actor at its last hidden expert layer, where each row of Φ is the weighted concatenation of expert features (Eq. (19)). With the expert-wise partition $\Phi = [\Phi_1 | \dots | \Phi_E]$, each block $\mathbf{A}_{\text{last},(e,e')}^{\text{exp}} = (1/N)\Phi_e^\top \Phi_{e'}$ measures routing-weighted within-expert correlations ($e=e'$) and cross-expert correlations ($e \neq e'$).

Writing $m = \sum_{e=1}^E d_{e,\text{last}}^{\text{exp}}$ and taking $I_m = \text{diag}(I_{d_{1,\text{last}}^{\text{exp}}}, \dots, I_{d_{E,\text{last}}^{\text{exp}}})$ to match the expert-wise block partition, the penalty decomposes as

$$\|\mathbf{A}_{\text{last}}^{\text{exp}} - \alpha I_m\|_F^2 = \sum_{e=1}^E \|\mathbf{A}_{\text{last},(e,e)}^{\text{exp}} - \alpha I_{d_{e,\text{last}}^{\text{exp}}}\|_F^2 + \sum_{e \neq e'} \|\mathbf{A}_{\text{last},(e,e')}^{\text{exp}}\|_F^2.$$

This follows from block additivity of the Frobenius norm: for any block matrix $B = [B_{e,e'}]$, $\|B\|_F^2 = \sum_{e,e'} \|B_{e,e'}\|_F^2$.

Within-expert effect. The diagonal-block term encourages each within-expert Gram $\mathbf{A}_{\text{last},(e,e)}^{\text{exp}}$ to be close to a scaled identity. Concretely, for $M = \mathbf{A}_{\text{last},(e,e)}^{\text{exp}} \in \mathbb{R}^{d \times d}$,

$$\|M - \alpha I_d\|_F^2 = \sum_{u \neq v} M_{uv}^2 + \sum_{u=1}^d (M_{uu} - \alpha)^2.$$

This follows by expanding the Frobenius norm entrywise, $\|X\|_F^2 = \sum_{u,v} X_{uv}^2$, and noting that $(M - \alpha I_d)_{uv} = M_{uv}$ for $u \neq v$ and $(M - \alpha I_d)_{uu} = M_{uu} - \alpha$. Here $M_{uv} = (1/N) \sum_i h_{i,e}^{(K)2} a_{e,u}^{\text{exp}}(x_i) a_{e,v}^{\text{exp}}(x_i)$ are routing-weighted cross-moments between feature dimensions, while M_{uu} are per-dimension second moments (variances if features are centered). Thus the penalty drives off-diagonal cross-moments toward zero and equalizes the diagonal entries, reducing within-expert feature correlations and balancing per-dimension scale under the routing weights.

Cross-expert effect. The off-diagonal-block term $\sum_{e \neq e'} \|\mathbf{A}_{\text{last},(e,e')}^{\text{exp}}\|_F^2$ penalizes cross-expert correlations, discouraging redundant representations shared across different experts.

Expanding the Frobenius norm gives the exact identity

$$\begin{aligned} \|\mathbf{A}_{\text{last},(e,e')}^{\text{exp}}\|_F^2 &= \frac{1}{N^2} \sum_{i=1}^N \sum_{j=1}^N h_{i,e}^{(K)} h_{i,e'}^{(K)} h_{j,e}^{(K)} h_{j,e'}^{(K)} \\ &\quad \cdot \langle a_e^{\text{exp}}(x_i), a_e^{\text{exp}}(x_j) \rangle \langle a_{e'}^{\text{exp}}(x_i), a_{e'}^{\text{exp}}(x_j) \rangle, \end{aligned}$$

Routing-weighted effect. Since $\mathbf{A}_{\text{last},(e,e')}^{\text{exp}}$ aggregates terms proportional to $h_{i,e}^{(K)} h_{i,e'}^{(K)} a_e^{\text{exp}}(x_i) a_{e'}^{\text{exp}}(x_i)^\top$, only states where experts are co-activated contribute to cross-expert coupling, and Top- K sparsity limits which expert pairs are coupled.

A.5. Training Hyperparameters

Table 2. Training hyperparameters for Top- K MoE PPO with SPHERE. Values summarize the default MetaWorld and HumanoidBench training setups.

	MetaWorld	HumanoidBench
Env steps per task	1,000,000	10,000,000
Vectorized envs (n_{env})	8	128
Observation normalization	none	VecNormalize
PPO rollout steps (n_{steps})	2048	16
PPO batch size	256	64
PPO epochs per update	10	10
Discount γ	0.99	0.99
GAE λ	0.95	0.95
Clip range	0.20	0.20
Max grad norm	0.50	0.50
Entropy coefficient	0.00	0.00
Value coefficient	0.50	0.50
Learning rate	linear $3\text{e-}4 \rightarrow 1\text{e-}4$	linear $3\text{e-}4 \rightarrow 1\text{e-}4$ over first half
Activation	ReLU	ReLU
MLP width	256	256
MLP depth	3	3
Number of experts E	10	10
Top- K	2	2
Gating softmax temperature τ	1.0	1.0
SPHERE ratio ρ	$1\text{e-}1$	$1\text{e-}3$

B. Matrix-Calculus Identities and Gradient Derivation

For completeness, we collect here the identities used in Section 4 and state the linear-layer gradient as a lemma.

Bias Absorption for Affine Layers. Throughout Section 4 we work with bias-free layers without loss of generality. Formally, let a generic gate or expert layer be affine, of the form $z_\ell = \mathbf{W}_\ell a_{\ell-1} + b_\ell$ with $\mathbf{W}_\ell \in \mathbb{R}^{d_\ell \times d_{\ell-1}}$, $a_{\ell-1} \in \mathbb{R}^{d_{\ell-1}}$, $b_\ell \in \mathbb{R}^{d_\ell}$, and $z_\ell \in \mathbb{R}^{d_\ell}$. By augmenting the activation with a constant

$$\tilde{a}_{\ell-1} = \begin{bmatrix} a_{\ell-1} \\ 1 \end{bmatrix}, \quad \tilde{\mathbf{W}}_\ell = [\mathbf{W}_\ell \ b_\ell],$$

we can rewrite the layer as a purely linear map

$$z_\ell = \tilde{\mathbf{W}}_\ell \tilde{a}_{\ell-1}.$$

Thus, after this reparameterization, each layer can be treated as bias-free by working with $\tilde{\mathbf{W}}_\ell$ and $\tilde{a}_{\ell-1}$. In Section 4 we therefore use the notation $\mathbf{W}_\ell$ and $a_{\ell-1}$ for this bias-absorbed form, and all layer widths $\{d_\ell\}$ are understood for this representation.

Gate and Expert Parameterization in Top- K MoE. Specializing to the Top- K MoE policy of Section 4, we write the full parameter vector as $\theta = (\psi, \{\theta_e\}_{e=1}^E)$, where ψ denotes the gate parameters and θ_e denotes the parameters of expert e . Let L^g and L^{exp} denote the numbers of gate and expert layers, respectively. After absorbing biases into the weights as above we take

$$\psi = \{\mathbf{W}_\ell^g\}_{\ell=1}^{L^g}, \quad \theta_e = \{\mathbf{W}_{e,\ell}^{\text{exp}}\}_{\ell=1}^{L^{\text{exp}}},$$

where $\mathbf{W}_\ell^g \in \mathbb{R}^{d_\ell^g \times d_{\ell-1}^g}$ are the gate weight matrices and $\mathbf{W}_{e,\ell}^{\text{exp}} \in \mathbb{R}^{d_{e,\ell}^{\text{exp}} \times d_{e,\ell-1}^{\text{exp}}}$ are the expert weight matrices for expert e at layer ℓ . Thus each gate or expert layer is treated as bias-free in the Jacobian Gram analysis, with parameters grouped into gate and expert blocks via $\{\mathbf{W}_\ell^g\}$ and $\{\mathbf{W}_{e,\ell}^{\text{exp}}\}$.

Lemma B.1 (Linear-layer gradient). *Let $z_\ell = \mathbf{W}_\ell a_{\ell-1}$ and $g_\ell = \partial f / \partial z_\ell$ for a scalar-valued function f . Then*

$$\frac{\partial f}{\partial \mathbf{W}_\ell} = g_\ell a_{\ell-1}^\top.$$

Proof. We use differential notation and consider the partial derivative of f with respect to W_ℓ , treating $a_{\ell-1}$ as independent of W_ℓ (i.e., $\partial a_{\ell-1} / \partial W_\ell = 0$). Then f depends on W_ℓ only through $z_\ell = W_\ell a_{\ell-1}$. Since f is scalar-valued in z_ℓ , its differential satisfies $df = (\partial f / \partial z_\ell)^\top dz_\ell = g_\ell^\top dz_\ell$. By the chain rule,

$$df = g_\ell^\top dz_\ell = g_\ell^\top d(W_\ell a_{\ell-1}) = g_\ell^\top dW_\ell a_{\ell-1}.$$

Using the scalar–trace identity $v^\top Xu = \text{Tr}(u v^\top X)$ for $v \in \mathbb{R}^m$, $X \in \mathbb{R}^{m \times n}$, and $u \in \mathbb{R}^n$ (see, e.g., Petersen & Pedersen, 2012), this can be written as

$$df = \text{Tr}(a_{\ell-1} g_\ell^\top dW_\ell) = \text{Tr}((g_\ell a_{\ell-1}^\top)^\top dW_\ell),$$

where the second equality uses cyclic invariance of trace, $\text{Tr}(ABC) = \text{Tr}(CAB)$. By the standard definition of the matrix derivative of a scalar function,

$$df = \text{Tr}((\partial f / \partial W_\ell)^\top dW_\ell) \quad \text{for all } dW_\ell,$$

which identifies $\partial f / \partial W_\ell = g_\ell a_{\ell-1}^\top$. $\square$

C. Top- K MoE Gradients and Block Khatri–Rao Structure

Overview. This appendix derives the expert-layer block Khatri–Rao structure used in Proposition 4.2 (Section 4). Along the way, it derives the same-layer gate and expert Jacobian Grams (Eqs. (32) and (37)), which are also used by later structural results (e.g., Appendix D).

C.1. Setup: Per-Sample Gradients

We first record the Top- K MoE per-sample gradients that will be used throughout the derivations below.

Top- K MoE Per-Sample Gradients. For each sample x_i , the scalar output can be written as

$$f(x_i) = \sum_{e' \in \mathcal{S}(x_i)} h_{i,e'}^{(K)} m_{e'}(x_i), \quad (27)$$

where $m_{e'}(x_i)$ is the contribution of expert e' . Let ϑ denote any one of the parameter blocks in θ ; for example, ϑ may be the gate parameters ψ , an expert block θ_e , or a specific weight matrix such as $\mathbf{W}_\ell^g$ or $\mathbf{W}_{e,\ell}^{\text{exp}}$. Differentiating $f(x_i)$ with respect to such a ϑ gives

$$\begin{aligned} \frac{\partial f(x_i)}{\partial \vartheta} &= \sum_{e' \in \mathcal{S}(x_i)} \frac{\partial}{\partial \vartheta} (h_{i,e'}^{(K)} m_{e'}(x_i)) \\ &= \sum_{e' \in \mathcal{S}(x_i)} \left(\frac{\partial h_{i,e'}^{(K)}}{\partial \vartheta} m_{e'}(x_i) + h_{i,e'}^{(K)} \frac{\partial m_{e'}(x_i)}{\partial \vartheta} \right). \end{aligned} \quad (28)$$

This is the *full* Top- K MoE gradient in block form.

C.2. Derivation of Same-Layer Jacobian Grams

In this section we derive the same-layer gate and expert Jacobian Grams starting from the per-sample Top- K MoE gradients in Eq. (28).

Gate Gradients. For the gate parameters ψ , the expert outputs $m_{e'}(x_i)$ do not depend on ψ , so the second term in Eq. (28) vanishes for all e' , giving

$$\frac{\partial f(x_i)}{\partial \psi} = \sum_{e' \in \mathcal{S}(x_i)} m_{e'}(x_i) \frac{\partial h_{i,e'}^{(K)}}{\partial \psi}.$$

At a specific gate layer ℓ , only the dependence of the gating weights $h_{i,e'}^{(K)}$ on the layer weights $\mathbf{W}_\ell^g$ contributes. Differentiating $f(x_i)$ with respect to $\mathbf{W}_\ell^g$ and using the product rule gives

$$\frac{\partial f(x_i)}{\partial \mathbf{W}_\ell^g} = \sum_{e' \in \mathcal{S}(x_i)} m_{e'}(x_i) \frac{\partial h_{i,e'}^{(K)}}{\partial \mathbf{W}_\ell^g}.$$

Writing the gate as a multilayer network with logits $s(x_i; \psi)$, pre-activations $z_\ell^g(x_i)$, and hidden activations $a_{\ell-1}^g(x_i)$, $h_{i,e'}^{(K)}$ depends on $\mathbf{W}_\ell^g$ only through $z_\ell^g(x_i)$. A second application of the chain rule gives

$$\frac{\partial h_{i,e'}^{(K)}}{\partial \mathbf{W}_\ell^g} = \frac{\partial h_{i,e'}^{(K)}}{\partial z_\ell^g(x_i)} \frac{\partial z_\ell^g(x_i)}{\partial \mathbf{W}_\ell^g}.$$

For the linear map $z_\ell^g(x_i) = \mathbf{W}_\ell^g a_{\ell-1}^g(x_i)$, applying Lemma B.1 with $f(x_i) = h_{i,e'}^{(K)}(x_i)$ and $g_\ell(x_i) = \partial h_{i,e'}^{(K)} / \partial z_\ell^g(x_i)$ gives

$$\frac{\partial h_{i,e'}^{(K)}}{\partial \mathbf{W}_\ell^g} = \frac{\partial h_{i,e'}^{(K)}}{\partial z_\ell^g(x_i)} (a_{\ell-1}^g(x_i))^\top,$$

so substituting into the previous expression for $\partial f(x_i) / \partial \mathbf{W}_\ell^g$ yields

$$\frac{\partial f(x_i)}{\partial \mathbf{W}_\ell^g} = \sum_{e' \in \mathcal{S}(x_i)} m_{e'}(x_i) \frac{\partial h_{i,e'}^{(K)}}{\partial z_\ell^g(x_i)} (a_{\ell-1}^g(x_i))^\top. \quad (29)$$

Thus gate gradients are explicitly weighted by the expert outputs. Differentiating Eq. (27) with respect to the gate pre-activation $z_\ell^g(x_i)$ and using that $m_{e'}(x_i)$ does not depend on gate activations gives

$$\frac{\partial f(x_i)}{\partial z_\ell^g(x_i)} = \sum_{e' \in \mathcal{S}(x_i)} m_{e'}(x_i) \frac{\partial h_{i,e'}^{(K)}}{\partial z_\ell^g(x_i)}.$$

We therefore identify the gate pre-activation gradient at layer ℓ as

$$g_\ell^g(x_i) := \frac{\partial f(x_i)}{\partial z_\ell^g(x_i)} = \sum_{e' \in \mathcal{S}(x_i)} m_{e'}(x_i) \frac{\partial h_{i,e'}^{(K)}}{\partial z_\ell^g(x_i)} \in \mathbb{R}^{d_\ell^g}. \quad (30)$$

and Eq. (29) can be written compactly as

$$\frac{\partial f(x_i)}{\partial \mathbf{W}_\ell^g} = g_\ell^g(x_i) (a_{\ell-1}^g(x_i))^\top, \quad (31)$$

where $g_\ell^g(x_i)$ is just a shorthand for the gate pre-activation gradient $\partial f(x_i) / \partial z_\ell^g(x_i)$ and still explicitly contains the expert-output weights $m_{e'}(x_i)$. Thus, like a standard dense layer, the gate layer admits a per-sample gradient in backprop–activation outer-product form. We define the gate-layer Jacobian block $\mathbf{J}_\ell^g$ to stack the vectorized per-sample gradients with respect to $\mathbf{W}_\ell^g$, so the i -th row is

$$j_{i,\ell}^{g^\top} := \left(\text{vec} \left(\frac{\partial f(x_i)}{\partial \mathbf{W}_\ell^g} \right) \right)^\top.$$

Using the standard vec–Kronecker identity $\text{vec}(uv^\top) = v \otimes u$ (see, e.g., Petersen & Pedersen, 2012; Horn & Johnson, 2012; Golub & Van Loan, 2013) on $\partial f(x_i) / \partial \mathbf{W}_\ell^g = g_\ell^g(x_i) (a_{\ell-1}^g(x_i))^\top$, we obtain

$$j_{i,\ell}^{g^\top} = (a_{\ell-1}^g(x_i) \otimes g_\ell^g(x_i))^\top.$$

Stacking these rows over $i=1, \dots, N$ gives the gate-layer Jacobian block $\mathbf{J}_\ell^g \in \mathbb{R}^{N \times p_\ell^g}$ with rows $j_{i,\ell}^{g^\top}$ (and $p_\ell^g = d_\ell^g d_{\ell-1}^g$). Its same-layer Gram expands as

$$\begin{aligned} (\mathbf{J}_\ell^g)^\top \mathbf{J}_\ell^g &= \sum_{i=1}^N j_{i,\ell}^{g^\top} (j_{i,\ell}^g)^\top \\ &= \sum_{i=1}^N (a_{\ell-1}^g(x_i) \otimes g_\ell^g(x_i)) (a_{\ell-1}^g(x_i) \otimes g_\ell^g(x_i))^\top \\ &= \sum_{i=1}^N (a_{\ell-1}^g(x_i) \otimes g_\ell^g(x_i)) (a_{\ell-1}^g(x_i)^\top \otimes g_\ell^g(x_i)^\top) \\ &= \sum_{i=1}^N (a_{\ell-1}^g(x_i) a_{\ell-1}^g(x_i)^\top) \otimes (g_\ell^g(x_i) g_\ell^g(x_i)^\top), \end{aligned} \quad (32)$$

where the last line uses the Kronecker product rule $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$ with $A = a_{\ell-1}^g(x_i)$, $B = g_\ell^g(x_i)$, $C = a_{\ell-1}^g(x_i)^\top$, and $D = g_\ell^g(x_i)^\top$.

Expert Gradients. For expert parameters $\vartheta \subset \theta_e$, differentiating Eq. (27) with respect to ϑ gives

$$\frac{\partial f(x_i)}{\partial \vartheta} = \sum_{e' \in S(x_i)} h_{i,e'}^{(K)} \frac{\partial m_{e'}(x_i)}{\partial \vartheta}.$$

Here the gate outputs $h_{i,e'}^{(K)}$ do not depend on ϑ , and $m_{e'}(x_i)$ depends on ϑ only when $e' = e$, so the sum reduces to

$$\frac{\partial f(x_i)}{\partial \vartheta} = h_{i,e}^{(K)} \frac{\partial m_e(x_i)}{\partial \vartheta}.$$

Specializing to the layer weights $\vartheta = \mathbf{W}_{e,\ell}^{\text{exp}}$ yields

$$\frac{\partial f(x_i)}{\partial \mathbf{W}_{e,\ell}^{\text{exp}}} = h_{i,e}^{(K)} \frac{\partial m_e(x_i)}{\partial \mathbf{W}_{e,\ell}^{\text{exp}}}. \quad (33)$$

Within expert e , we write the layer- ℓ pre-activation on sample x_i as

$$z_{e,\ell}(x_i) = \mathbf{W}_{e,\ell}^{\text{exp}} a_{e,\ell-1}^{\text{exp}}(x_i), \quad a_{e,\ell-1}^{\text{exp}}(x_i) \in \mathbb{R}^{d_{e,\ell-1}^{\text{exp}}},$$

and define the corresponding backprop signal

$$g_{e,\ell}^{\text{exp}}(x_i) := \frac{\partial m_e(x_i)}{\partial z_{e,\ell}(x_i)} \in \mathbb{R}^{d_{e,\ell}^{\text{exp}}}.$$

By the chain rule and the linear-layer gradient Lemma B.1 applied to the scalar function $m_e(x_i)$ and the map $z_{e,\ell}(x_i) = \mathbf{W}_{e,\ell}^{\text{exp}} a_{e,\ell-1}^{\text{exp}}(x_i)$,

$$\frac{\partial m_e(x_i)}{\partial \mathbf{W}_{e,\ell}^{\text{exp}}} = g_{e,\ell}^{\text{exp}}(x_i) a_{e,\ell-1}^{\text{exp}}(x_i)^\top. \quad (34)$$

Substituting into the previous expression yields

$$\frac{\partial f(x_i)}{\partial \mathbf{W}_{e,\ell}^{\text{exp}}} = h_{i,e}^{(K)} g_{e,\ell}^{\text{exp}}(x_i) a_{e,\ell-1}^{\text{exp}}(x_i)^\top = g_{e,\ell}^{\text{exp}}(x_i) (h_{i,e}^{(K)} a_{e,\ell-1}^{\text{exp}}(x_i))^\top. \quad (35)$$

Vectorizing and transposing gives the gradient row

$$j_{i,e,\ell}^\top := \left(\text{vec}(g_{e,\ell}^{\text{exp}}(x_i) (h_{i,e}^{(K)} a_{e,\ell-1}^{\text{exp}}(x_i))^\top) \right)^\top = (h_{i,e}^{(K)} a_{e,\ell-1}^{\text{exp}}(x_i) \otimes g_{e,\ell}^{\text{exp}}(x_i))^\top = h_{i,e}^{(K)} (a_{e,\ell-1}^{\text{exp}}(x_i) \otimes g_{e,\ell}^{\text{exp}}(x_i))^\top, \quad (36)$$

where we used the same vec-Kronecker identity $\text{vec}(uv^\top) = v \otimes u$ and the fact that the scalar $h_{i,e}^{(K)}$ can be absorbed into either factor of the Kronecker product.

Let $\mathbf{J}_{e,\ell}^{\text{exp}} \in \mathbb{R}^{N \times p_{e,\ell}^{\text{exp}}}$ collect the Jacobian rows for expert e at layer ℓ , so its i -th row is $j_{i,e,\ell}^\top$, and stack experts within layer ℓ as $\mathbf{J}_\ell^{\text{exp}} = [\mathbf{J}_{1,\ell}^{\text{exp}}, \dots, \mathbf{J}_{E,\ell}^{\text{exp}}] \in \mathbb{R}^{N \times p_\ell^{\text{exp}}}$ with $p_\ell^{\text{exp}} = \sum_e p_{e,\ell}^{\text{exp}}$ and rows $j_{i,\ell}^\top$. The exact same-layer Gram $(\mathbf{J}_\ell^{\text{exp}})^\top \mathbf{J}_\ell^{\text{exp}}$ is then an $E \times E$ block matrix

$$(\mathbf{J}_\ell^{\text{exp}})^\top \mathbf{J}_\ell^{\text{exp}} = \begin{bmatrix} (\mathbf{J}_{1,\ell}^{\text{exp}})^\top \mathbf{J}_{1,\ell}^{\text{exp}} & \dots & (\mathbf{J}_{1,\ell}^{\text{exp}})^\top \mathbf{J}_{E,\ell}^{\text{exp}} \\ \vdots & \ddots & \vdots \\ (\mathbf{J}_{E,\ell}^{\text{exp}})^\top \mathbf{J}_{1,\ell}^{\text{exp}} & \dots & (\mathbf{J}_{E,\ell}^{\text{exp}})^\top \mathbf{J}_{E,\ell}^{\text{exp}} \end{bmatrix},$$

whose (e, e') block is

$$(\mathbf{J}_{e,\ell}^{\text{exp}})^\top \mathbf{J}_{e',\ell}^{\text{exp}} = \sum_{i=1}^N j_{i,e,\ell}^\top j_{i,e',\ell}.$$

Substituting the per-sample form $j_{i,e,\ell} = h_{i,e}^{(K)} \text{vec}(g_{e,\ell}^{\text{exp}}(x_i) a_{e,\ell-1}^{\text{exp}}(x_i)^\top)$ and using $\text{vec}(uv^\top) = v \otimes u$ gives $j_{i,e,\ell} = h_{i,e}^{(K)} (a_{e,\ell-1}^{\text{exp}}(x_i) \otimes g_{e,\ell}^{\text{exp}}(x_i))$, so

$$\begin{aligned} (\mathbf{J}_{e,\ell}^{\text{exp}})^\top \mathbf{J}_{e',\ell}^{\text{exp}} &= \sum_{i=1}^N j_{i,e,\ell} j_{i,e',\ell}^\top \\ &= \sum_{i=1}^N h_{i,e}^{(K)} h_{i,e'}^{(K)} (a_{e,\ell-1}^{\text{exp}}(x_i) \otimes g_{e,\ell}^{\text{exp}}(x_i)) (a_{e',\ell-1}^{\text{exp}}(x_i) \otimes g_{e',\ell}^{\text{exp}}(x_i))^\top \\ &= \sum_{i=1}^N h_{i,e}^{(K)} h_{i,e'}^{(K)} (a_{e,\ell-1}^{\text{exp}}(x_i) \otimes g_{e,\ell}^{\text{exp}}(x_i)) (a_{e',\ell-1}^{\text{exp}}(x_i)^\top \otimes g_{e',\ell}^{\text{exp}}(x_i)^\top) \\ &= \sum_{i=1}^N h_{i,e}^{(K)} h_{i,e'}^{(K)} (a_{e,\ell-1}^{\text{exp}}(x_i) a_{e',\ell-1}^{\text{exp}}(x_i)^\top) \otimes (g_{e,\ell}^{\text{exp}}(x_i) g_{e',\ell}^{\text{exp}}(x_i)^\top), \end{aligned} \quad (37)$$

where the last line uses the Kronecker rule $(A \otimes B)(C \otimes D) = (AC) \otimes (BD)$. The within-layer independence approximation step and the resulting block Khatri–Rao form are collected in Appendix C.3.

C.3. Within-Layer Independence and Block Khatri–Rao Form

We summarize how the exact expert-layer Jacobian Grams in Eq. (37) lead to the block-factorized form used in Section 4 and in Proposition 4.2.

Collecting the blocks $\{(\mathbf{J}_{e,\ell}^{\text{exp}})^\top \mathbf{J}_{e',\ell}^{\text{exp}}\}_{e,e'}$ gives the exact same-layer expert Gram $(\mathbf{J}_\ell^{\text{exp}})^\top \mathbf{J}_\ell^{\text{exp}}$ as an $E \times E$ block matrix whose (e, e') block is a *sum* of Kronecker products between an activation factor $a_{e,\ell-1}^{\text{exp}}(x_i) a_{e',\ell-1}^{\text{exp}}(x_i)^\top$ and a gradient factor $g_{e,\ell}^{\text{exp}}(x_i) g_{e',\ell}^{\text{exp}}(x_i)^\top$, cf. Eq. (37). For later use it is convenient to introduce the corresponding empirical activation and gradient Grams at the expert layer. Define

$$\begin{aligned} \mathbf{A}_{\ell-1,(e,e')}^{\text{exp}} &= \frac{1}{N} \sum_{i=1}^N h_{i,e}^{(K)} h_{i,e'}^{(K)} a_{e,\ell-1}^{\text{exp}}(x_i) a_{e',\ell-1}^{\text{exp}}(x_i)^\top, \\ \mathbf{G}_{\ell,(e,e')}^{\text{exp}} &= \frac{1}{N} \sum_{i=1}^N g_{e,\ell}^{\text{exp}}(x_i) g_{e',\ell}^{\text{exp}}(x_i)^\top, \end{aligned} \quad (38)$$

and collect them into block matrices $\mathbf{A}_{\ell-1}^{\text{exp}} = [\mathbf{A}_{\ell-1,(e,e')}^{\text{exp}}]_{e,e'}$ and $\mathbf{G}_\ell^{\text{exp}} = [\mathbf{G}_{\ell,(e,e')}^{\text{exp}}]_{e,e'}$. In terms of these empirical Grams, the *normalized* same-layer expert Gram

$$\mathbf{G}_\ell^{\text{GN,exp}} = \frac{1}{N} (\mathbf{J}_\ell^{\text{exp}})^\top \mathbf{J}_\ell^{\text{exp}}$$

has (e, e') block

$$\begin{aligned} \mathbf{G}_{e,e',\ell}^{\text{GN,exp}} &= \frac{1}{N} (\mathbf{J}_{e,\ell}^{\text{exp}})^\top \mathbf{J}_{e',\ell}^{\text{exp}} \\ &= \frac{1}{N} \sum_{i=1}^N h_{i,e}^{(K)} h_{i,e'}^{(K)} (a_{e,\ell-1}^{\text{exp}}(x_i) a_{e',\ell-1}^{\text{exp}}(x_i)^\top) \otimes (g_{e,\ell}^{\text{exp}}(x_i) g_{e',\ell}^{\text{exp}}(x_i)^\top). \end{aligned}$$

Viewing the minibatch index i as a draw from the empirical data distribution, the sum above is an empirical estimate of a fourth-order moment in the activation and gradient factors, since each entry involves terms of the form $\mathbb{E}[a_u a_v g_p g_q]$. A standard within-layer independence approximation between activations and backprop gradients (see, e.g., [Martens & Grosse, 2015](#); [Grosse & Martens, 2016](#)) factorizes this fourth-order moment into a product of second moments, yielding $\mathbb{E}[a_u a_v g_p g_q] \approx \mathbb{E}[a_u a_v] \mathbb{E}[g_p g_q]$; substituting empirical averages for expectations yields the block-factorized approximation

$$\mathbf{G}_{e,e',\ell}^{\text{GN,exp}} \approx \mathbf{A}_{\ell-1,(e,e')}^{\text{exp}} \otimes \mathbf{G}_{\ell,(e,e')}^{\text{exp}}.$$

Collecting these approximated blocks over all (e, e') pairs, the expert-layer Gram can be written compactly as a block Khatri–Rao product

$$\begin{aligned} \mathbf{G}_\ell^{\text{GN,exp}} &\approx \mathbf{A}_{\ell-1}^{\text{exp}} * \mathbf{G}_\ell^{\text{exp}}, \\ [\mathbf{A}_{\ell-1}^{\text{exp}} * \mathbf{G}_\ell^{\text{exp}}]_{e,e'} &= \mathbf{A}_{\ell-1,(e,e')}^{\text{exp}} \otimes \mathbf{G}_{\ell,(e,e')}^{\text{exp}}, \end{aligned}$$

where $*$ denotes the block Khatri–Rao product on corresponding blocks.

To make the activation blocks explicit, for an expert layer ℓ we define the *concatenated expert feature* for sample i as

$$a_{\ell-1}^{\text{ce}}(x_i) = \begin{bmatrix} h_{i,1}^{(K)} a_{1,\ell-1}^{\text{exp}}(x_i)^\top & | & h_{i,2}^{(K)} a_{2,\ell-1}^{\text{exp}}(x_i)^\top & | & \cdots \\ & & & & | & h_{i,E}^{(K)} a_{E,\ell-1}^{\text{exp}}(x_i)^\top \end{bmatrix}^\top. \quad (39)$$

This vector lies in $\mathbb{R}^{D_{\ell-1}}$, where $D_{\ell-1} = \sum_{e=1}^E d_{e,\ell-1}^{\text{exp}}$. Stacking over the minibatch gives the concatenated expert feature matrix $\Phi_{\ell-1}^{\text{ce}} \in \mathbb{R}^{N \times D_{\ell-1}}$ with rows $(a_{\ell-1}^{\text{ce}}(x_i))^\top$. The concatenated expert feature Gram is

$$\mathbf{A}_{\ell-1}^{\text{ce}} = \frac{1}{N} (\Phi_{\ell-1}^{\text{ce}})^\top \Phi_{\ell-1}^{\text{ce}} \in \mathbb{R}^{D_{\ell-1} \times D_{\ell-1}},$$

with (e, e') block

$$[\mathbf{A}_{\ell-1}^{\text{ce}}]_{e,e'} = \frac{1}{N} \sum_{i=1}^N h_{i,e}^{(K)} h_{i,e'}^{(K)} a_{e,\ell-1}^{\text{exp}}(x_i) a_{e',\ell-1}^{\text{exp}}(x_i)^\top.$$

Thus the empirical block activation Gram $\mathbf{A}_{\ell-1}^{\text{exp}}$ above coincides with the concatenated expert feature Gram: $\mathbf{A}_{\ell-1}^{\text{exp}} = \mathbf{A}_{\ell-1}^{\text{ce}}$. For notational convenience, for expert layers we also define the concatenated expert gradients

$$g_\ell^{\text{exp}}(x_i) := [g_{1,\ell}^{\text{exp}}(x_i)^\top \mid \cdots \mid g_{E,\ell}^{\text{exp}}(x_i)^\top]^\top,$$

so that $\mathbf{G}_\ell^{\text{exp}} = (1/N) \sum_i g_\ell^{\text{exp}}(x_i) g_\ell^{\text{exp}}(x_i)^\top$ contains the within- and cross-expert gradient covariances at layer ℓ .

D. Tracy–Singh Factorization and Isotropy Transfer

Here we collect the structural results used to relate the expert-layer Khatri–Rao Grams to their Kronecker proxies and to transfer spectral isotropy from the proxy to the expert-layer Gram.

Proposition D.1 (Expert block Khatri–Rao as a Tracy–Singh principal submatrix with Kronecker spectrum). *Let $\mathbf{A} = \mathbf{A}_{\ell-1}^{\text{ce}} \in \mathbb{R}^{D_{\ell-1} \times D_{\ell-1}}$ and $\mathbf{G} = \mathbf{G}_\ell^{\text{exp}} \in \mathbb{R}^{d_\ell \times d_\ell}$ be the concatenated expert feature and gradient Grams at expert layer ℓ defined in Section 4, with symmetric expert-wise block partitions $\{\mathbf{A}_{e,e'}\}$ and $\{\mathbf{G}_{e,e'}\}$. Denote by $\mathbf{T}_\ell := \mathbf{A} \boxtimes \mathbf{G}$ the Tracy–Singh product of $\mathbf{A}$ and $\mathbf{G}$, and by $\mathbf{K}_\ell := \mathbf{A} \otimes \mathbf{G}$ the Kronecker proxy. Then:*

1. $\mathbf{T}_\ell$ is SPSP and admits a Gram factorization $\mathbf{T}_\ell = \mathbf{Z}_\ell \mathbf{Z}_\ell^\top$ for a suitable matrix $\mathbf{Z}_\ell$ constructed from blockwise Cholesky factors of $\mathbf{A}$ and $\mathbf{G}$.
2. The expert-layer Gram equals a Khatri–Rao block of $\mathbf{T}_\ell$ and can be written as a principal submatrix

$$\mathbf{G}_\ell^{\text{GN,exp}} = \mathbf{A} * \mathbf{G} = \mathbf{Z}_\ell'^\top \mathbf{T}_\ell \mathbf{Z}_\ell',$$

for some column-orthonormal selection matrix $\mathbf{Z}_\ell'$ (so that $\mathbf{Z}_\ell'^\top \mathbf{Z}_\ell' = \mathbf{I}$) that selects expert-aligned blocks.

3. $\mathbf{T}_\ell$ and the Kronecker proxy $\mathbf{K}_\ell$ share the same multiset of eigenvalues, and hence the same spectral-entropy effective rank:

$$\{\lambda(\mathbf{T}_\ell)\} \equiv \{\lambda(\mathbf{K}_\ell)\}, \quad r_e(\mathbf{T}_\ell) = r_e(\mathbf{K}_\ell).$$

Proof. For (i), PSD of $\mathbf{A}$ and $\mathbf{G}$ implies the existence of factorizations $\mathbf{A} = \mathbf{X} \mathbf{X}^\top$ and $\mathbf{G} = \mathbf{Y} \mathbf{Y}^\top$ with $\mathbf{X} = [\mathbf{X}_1^\top, \dots, \mathbf{X}_E^\top]^\top$ and $\mathbf{Y} = [\mathbf{Y}_1^\top, \dots, \mathbf{Y}_E^\top]^\top$ block-partitioned compatibly with the experts, so $\mathbf{A}_{e,e'} = \mathbf{X}_e \mathbf{X}_{e'}^\top$ and $\mathbf{G}_{e,e'} = \mathbf{Y}_e \mathbf{Y}_{e'}^\top$. By the definition of the Tracy–Singh product, the (e, e') block of $\mathbf{T}_\ell = \mathbf{A} \boxtimes \mathbf{G}$ is

$$\mathbf{T}_\ell^{(e,e')} = \mathbf{A}_{e,e'} \otimes \mathbf{G}_{e,e'} = (\mathbf{X}_e \mathbf{X}_{e'}^\top) \otimes (\mathbf{Y}_e \mathbf{Y}_{e'}^\top) = (\mathbf{X}_e \otimes \mathbf{Y}_e)(\mathbf{X}_{e'} \otimes \mathbf{Y}_{e'})^\top.$$

Stacking the block rows $\mathbf{Z}_e := \mathbf{X}_e \otimes \mathbf{Y}_e$ as $\mathbf{Z}_\ell^\top := [\mathbf{Z}_1^\top, \dots, \mathbf{Z}_E^\top]$ yields

$$\mathbf{T}_\ell = \mathbf{Z}_\ell \mathbf{Z}_\ell^\top,$$

which is symmetric PSD by construction.

For (ii), Liu & Trenkler (2008, Section 2.2) show that, under the same symmetric block partition, the (block) Khatri–Rao product can be obtained from the Tracy–Singh product by a single selection matrix Z'_ℓ satisfying $Z'_\ell{}^\top Z'_\ell = I$, namely

$$\mathbf{A} * \mathbf{G} = Z'_\ell{}^\top \mathbf{T}_\ell Z'_\ell.$$

By Eq. (37), $\mathbf{G}_\ell^{\text{GN}, \text{exp}}$ has exactly this block Khatri–Rao structure, so $\mathbf{G}_\ell^{\text{GN}, \text{exp}} = \mathbf{A} * \mathbf{G}$ and is therefore a principal submatrix of $\mathbf{T}_\ell$ selected on expert-aligned blocks.

For (iii), Lemma 3 in Liu & Trenkler (2008) states that the Tracy–Singh product is related to the Kronecker product via orthogonal permutations:

$$\mathbf{T}_\ell = P_1^\top \mathbf{K}_\ell P_2, \quad P_1^\top P_1 = P_2^\top P_2 = I.$$

Thus $\mathbf{T}_\ell$ and $\mathbf{K}_\ell$ have the same singular values. Since both are symmetric PSD (the latter because the Kronecker of PSDs is PSD), singular values and eigenvalues coincide for each, so their eigenvalue multisets, and hence their spectral-entropy effective ranks, are identical. $\square$

Thus the Kronecker proxy $\mathbf{K}_\ell = \mathbf{A}_{\ell-1}^{\text{ce}} \otimes \mathbf{G}_\ell^{\text{exp}}$ provides a PSD spectral envelope for the expert-layer Gram $\mathbf{G}_\ell^{\text{GN}, \text{exp}}$ via the Tracy–Singh factorization $Z_\ell Z_\ell^\top$ and the principal-submatrix relation in Proposition D.1. In particular, any spectral isotropy (scaled-identity structure) imposed on $\mathbf{K}_\ell$ immediately transfers to $\mathbf{T}_\ell$, and Cauchy interlacing for principal submatrices (Proposition D.2 below) then implies that the eigenvalues of $\mathbf{G}_\ell^{\text{GN}, \text{exp}}$ are driven toward equality as well. Since PSD matrices attain maximal spectral-entropy effective rank exactly at scaled identities (Proposition E.3), regularizing $\mathbf{A}_{\ell-1}^{\text{ce}} \otimes \mathbf{G}_\ell^{\text{exp}}$ toward isotropy maximizes the effective rank of both the Tracy–Singh envelope and its expert-layer principal submatrix.

Proposition D.2 (Isotropy transfer and effective rank). *Let $M \in \mathbb{R}^{n \times n}$ be SPSP with eigenvalues $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ and let $B \in \mathbb{R}^{k \times k}$ be a principal submatrix of M (equivalently $B = P^\top M P$ for some $P \in \mathbb{R}^{n \times k}$ with $P^\top P = I_k$). Denote the spectral spread $\Delta(M) = \lambda_1 - \lambda_n$ and $\Delta(B) = \mu_1 - \mu_k$ where $\mu_1 \geq \dots \geq \mu_k$ are the eigenvalues of B . Then $\Delta(B) \leq \Delta(M)$. Moreover, if a sequence of PSD matrices $\{M_t\}$ satisfies $\Delta(M_t) \rightarrow 0$ and we write $B_t = P^\top M_t P$, then $\Delta(B_t) \rightarrow 0$ and every nonzero eigenvalue of B_t converges to a common limit.*

Proof. By Cauchy interlacing for eigenvalues of principal submatrices (see, e.g., Horn & Johnson, 2012; Golub & Van Loan, 2013), if M has ordered eigenvalues $\lambda_1 \geq \dots \geq \lambda_n$ and B is any $k \times k$ principal submatrix with ordered eigenvalues $\mu_1 \geq \dots \geq \mu_k$, then

$$\lambda_i \geq \mu_i \geq \lambda_{i+n-k} \quad \text{for } i = 1, \dots, k.$$

In particular, for $i=1$ and $i=k$ this gives

$$\lambda_1 \geq \mu_1, \quad \mu_k \geq \lambda_n,$$

and hence $\Delta(B) = \mu_1 - \mu_k \leq \lambda_1 - \lambda_n = \Delta(M)$, which is the first claim.

For the second claim, let $\{M_t\}$ be a sequence of PSD matrices with eigenvalues $\{\lambda_i^{(t)}\}_{i=1}^n$ such that $\Delta(M_t) = \lambda_1^{(t)} - \lambda_n^{(t)} \rightarrow 0$. Fix a principal-submatrix pattern P and define $B_t = P^\top M_t P$ with eigenvalues $\{\mu_i^{(t)}\}_{i=1}^k$. By interlacing applied to each t ,

$$\lambda_1^{(t)} \geq \mu_i^{(t)} \geq \lambda_n^{(t)} \quad \text{for all } i = 1, \dots, k.$$

Since $\lambda_1^{(t)} - \lambda_n^{(t)} \rightarrow 0$, the squeeze theorem implies that every $\mu_i^{(t)}$ converges to the common limit of $\lambda_1^{(t)}$ and $\lambda_n^{(t)}$, and in particular $\Delta(B_t) = \mu_1^{(t)} - \mu_k^{(t)} \rightarrow 0$. For any nonzero eigenvalue subsequence this common limit is strictly positive, so asymptotically all nonzero eigenvalues of B_t coincide. $\square$

E. Effective-Rank Identities and Proofs

Lemma E.1 (Kronecker effective rank). *Let A and B have nonzero singular values $\{a_i\}_i$ and $\{b_j\}_j$ with $z = \sum_i a_i$ and $w = \sum_j b_j$. Then*

$$r_e(A \otimes B) = r_e(A) r_e(B).$$

Proof. Singular values of $A \otimes B$ are $\{a_i b_j\}_{i,j}$ (see, e.g., Horn & Johnson, 2012; Golub & Van Loan, 2013). Normalizing,

$$p_{ij} := \frac{a_i b_j}{\sum_{i',j'} a_{i'} b_{j'}} = \frac{a_i}{z} \cdot \frac{b_j}{w} = p_i^A p_j^B.$$

Hence $H(p^{A \otimes B}) = -\sum_{i,j} p_{ij} \log p_{ij} = -\sum_i p_i^A \log p_i^A - \sum_j p_j^B \log p_j^B = H(p^A) + H(p^B)$. Exponentiating gives the claim. $\square$

Corollary E.1.1 (Kronecker block effective rank). *If $\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}} = \mathbf{A}_\ell^{\text{exp}} \otimes \mathbf{G}_\ell^{\text{exp}}$, then $r_e(\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}}) = r_e(\mathbf{A}_\ell^{\text{exp}}) r_e(\mathbf{G}_\ell^{\text{exp}})$.*

Proof. Apply Lemma E.1 with $A = \mathbf{A}_\ell^{\text{exp}}$ and $B = \mathbf{G}_\ell^{\text{exp}}$. $\square$

Corollary E.1.2 (Monotonicity under the Kronecker proxy). *Assume $\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}} = \mathbf{A}_\ell^{\text{exp}} \otimes \mathbf{G}_\ell^{\text{exp}}$. Then, for fixed $\mathbf{G}_\ell^{\text{exp}}$, $r_e(\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}})$ strictly increases with $r_e(\mathbf{A}_\ell^{\text{exp}})$.*

Proof. By Lemma E.1, $r_e(\tilde{\mathbf{G}}_\ell^{\text{GN},\text{exp}}) = r_e(\mathbf{A}_\ell^{\text{exp}}) r_e(\mathbf{G}_\ell^{\text{exp}})$; fixing $r_e(\mathbf{G}_\ell^{\text{exp}})$ makes the product strictly increase with $r_e(\mathbf{A}_\ell^{\text{exp}})$. $\square$

Lemma E.2 (Block-diagonal effective rank). *Let $G = \bigoplus_{\ell=1}^L G_\ell$ be block-diagonal and denote the multiset of singular values of G_ℓ by $\{\sigma_{\ell,i}\}_i$. For each block, set $s_\ell = \sum_i \sigma_{\ell,i}$ (the nuclear norm $\|G_\ell\|_*$) and assume $\sum_\ell s_\ell > 0$. Define mixture weights*

$$\alpha_\ell := \frac{s_\ell}{\sum_m s_m},$$

so that $\sum_\ell \alpha_\ell = 1$. For every block with $s_\ell > 0$, let $r_e(G_\ell)$ denote its spectral-entropy effective rank as in Eq. (6), computed from the normalized singular-value spectrum of G_ℓ . Then the spectral-entropy effective rank of G satisfies

$$r_e(G) = \exp\left(H(\alpha) + \sum_{\{\ell: s_\ell > 0\}} \alpha_\ell \log r_e(G_\ell)\right),$$

where $H(\alpha) = -\sum_\ell \alpha_\ell \log \alpha_\ell$ is the Shannon entropy of the mixture weights.

Proof. By the direct-sum spectral property (e.g., [Horn & Johnson, 2012](#); [Golub & Van Loan, 2013](#)), the multiset of singular values of G equals the multiset union of those of the blocks:

$$\{\sigma(G)\} \equiv \biguplus_{\ell=1}^L \{\sigma_{\ell,i}\}_i.$$

Let $s_\ell = \sum_i \sigma_{\ell,i}$ and $s = \sum_m s_m$; by assumption $s > 0$. For each block define $\alpha_\ell = s_\ell/s$ and, when $s_\ell > 0$, the within-block normalized spectrum $q_{\ell,i} = \sigma_{\ell,i}/s_\ell$ (for $s_\ell = 0$, set $q_{\ell,i} = 0$). Then the globally normalized singular-value distribution is

$$p_{\ell,i} = \frac{\sigma_{\ell,i}}{\sum_{k,j} \sigma_{k,j}} = \frac{\sigma_{\ell,i}}{s} = \frac{s_\ell}{s} \frac{\sigma_{\ell,i}}{s_\ell} = \alpha_\ell q_{\ell,i} \quad (\text{for } s_\ell > 0),$$

with $\sum_\ell \alpha_\ell = 1$ and, for all ℓ with $s_\ell > 0$, $\sum_i q_{\ell,i} = 1$. The Shannon entropy of p decomposes as

$$\begin{aligned} H(p) &:= -\sum_{\ell,i} p_{\ell,i} \log p_{\ell,i} = -\sum_{\{\ell: s_\ell > 0\}, i} \alpha_\ell q_{\ell,i} (\log \alpha_\ell + \log q_{\ell,i}) \\ &= -\sum_{\{\ell: s_\ell > 0\}} \alpha_\ell \log \alpha_\ell \sum_i q_{\ell,i} - \sum_{\{\ell: s_\ell > 0\}} \alpha_\ell \sum_i q_{\ell,i} \log q_{\ell,i} \\ &= H(\alpha) + \sum_{\{\ell: s_\ell > 0\}} \alpha_\ell H(q_\ell), \end{aligned}$$

where $H(\alpha) = -\sum_\ell \alpha_\ell \log \alpha_\ell$ and $H(q_\ell) := -\sum_i q_{\ell,i} \log q_{\ell,i}$ is the entropy of the normalized spectrum in block ℓ . By definition $r_e(G_\ell) = \exp(H(q_\ell))$ for blocks with $s_\ell > 0$, so exponentiating $H(p)$ yields

$$r_e(G) = \exp\left(H(\alpha) + \sum_{\{\ell: s_\ell > 0\}} \alpha_\ell \log r_e(G_\ell)\right),$$

which is the claimed expression. $\square$

Proposition E.3. Let $M \in \mathbb{R}^{m \times m}$ be positive semidefinite. Then $r_e(M) \leq m$, with equality iff $M = \alpha \mathbf{I}_m$ for some $\alpha > 0$ (i.e., all eigenvalues are equal).

Proof. Let $\{\sigma_i(M)\}_{i=1}^m$ be the singular values and set $r := |\{i : \sigma_i(M) > 0\}| = \text{rank}(M)$. For indices with $\sigma_i > 0$, define $p_i := \sigma_i / \sum_{j: \sigma_j > 0} \sigma_j$. Then

$$r_e(M) = \exp \left(- \sum_{\{i: \sigma_i > 0\}} p_i \log p_i \right) \leq r,$$

with equality iff $p_i \equiv 1/r$ for all i (Schur-concavity of entropy; [Marshall et al., 2011](#)), i.e., the nonzero singular values are equal. In particular, $r_e(M) = m$ iff $r = m$ (full rank) and $\sigma_1 = \dots = \sigma_m$. Writing the symmetric SVD $M = U \Sigma U^\top$ (PSD implies $U = V$), if $\Sigma = \alpha \mathbf{I}_m$ then $M = U \Sigma U^\top = \alpha \mathbf{I}_m$. $\square$

Proof of Proposition 4.4. By Corollary E.1.1, $r_e(\tilde{\mathbf{G}}_\ell^{\text{GN,exp}}) = r_e(\mathbf{A}_\ell^{\text{exp}}) r_e(\mathbf{G}_\ell^{\text{exp}})$, hence with $\mathbf{G}_\ell^{\text{exp}}$ fixed each $r_e(\tilde{\mathbf{G}}_\ell^{\text{GN,exp}})$ strictly increases with $r_e(\mathbf{A}_\ell^{\text{exp}})$. Under the block-diagonal model, Lemma E.2 gives

$$\log r_e(\tilde{\mathbf{G}}^{\text{GN,exp}}) = H(\alpha) + \sum_{\{\ell: s_\ell > 0\}} \alpha_\ell \log r_e(\tilde{\mathbf{G}}_\ell^{\text{GN,exp}}),$$

with mixture weights α_ℓ held fixed by assumption. This is an affine function of the block entropies $\log r_e(\tilde{\mathbf{G}}_\ell^{\text{GN,exp}})$ with nonnegative coefficients α_ℓ that sum to one. Therefore, increasing any $\log r_e(\tilde{\mathbf{G}}_\ell^{\text{GN,exp}})$ (and hence any $r_e(\tilde{\mathbf{G}}_\ell^{\text{GN,exp}})$) strictly increases $\log r_e(\tilde{\mathbf{G}}^{\text{GN,exp}})$ and thus $r_e(\tilde{\mathbf{G}}^{\text{GN,exp}})$. Combining with the previous step yields the claimed strict monotonicity of $r_e(\tilde{\mathbf{G}}^{\text{GN,exp}})$ in each $r_e(\mathbf{A}_\ell^{\text{exp}})$. $\square$

Proof of Proposition 4.5. Each matrix $\mathbf{A}_\ell^{\text{exp}}$ is PSD. Applying Proposition E.3 with $M = \mathbf{A}_\ell^{\text{exp}}$ and dimension $m = d_{\ell-1}$ yields the stated upper bound on $r_e(\mathbf{A}_\ell^{\text{exp}})$ and the characterization of the maximizers as scaled identities. $\square$

F. Perturbation Bounds for Approximation Errors

Effective-rank stability under matrix perturbations. We control approximation error by bounding the total-variation distance between the normalized spectra of two PSD matrices, which directly upper bounds the change in spectral entropy (hence $\log r_e$). *Remark.* If the perturbation is small in Frobenius norm relative to the trace scale, then $r_e(\cdot)$ is stable up to a multiplicative factor that grows smoothly with this ratio. These worst-case continuity bounds can be conservative; we use them to make explicit which coupling terms must be small, and rely on the empirical diagnostics in Figures 8 and 11 to assess their magnitude in practice.

Lemma F.1 (Stability of r_e under Frobenius perturbations). Let $K, \tilde{K} \in \mathbb{R}^{N \times N}$ be symmetric PSD with $\text{Tr}(K) > 0$ and $\text{Tr}(\tilde{K}) > 0$. Let $\lambda(K) \in \mathbb{R}_+^N$ denote the vector of eigenvalues (with multiplicity) and define normalized spectra

$$p := \frac{\lambda(K)}{\text{Tr}(K)}, \quad \tilde{p} := \frac{\lambda(\tilde{K})}{\text{Tr}(\tilde{K})}.$$

Let $\tau := \min\{\text{Tr}(K), \text{Tr}(\tilde{K})\}$ and $\Delta := K - \tilde{K}$. Define

$$\varepsilon := \frac{\sqrt{N} \|\Delta\|_F}{\tau}, \quad h(\varepsilon) := -\varepsilon \log \varepsilon - (1 - \varepsilon) \log(1 - \varepsilon).$$

If $\varepsilon \leq 1 - 1/N$, then

$$|\log r_e(K) - \log r_e(\tilde{K})| = |H(p) - H(\tilde{p})| \leq \varepsilon \log(N - 1) + h(\varepsilon),$$

and hence

$$e^{-(\varepsilon \log(N-1) + h(\varepsilon))} \leq \frac{r_e(K)}{r_e(\tilde{K})} \leq e^{\varepsilon \log(N-1) + h(\varepsilon)}.$$

Proof. Hoffman–Wielandt gives $\|\lambda(K) - \lambda(\tilde{K})\|_2 \leq \|\Delta\|_F$, hence $\|\lambda(K) - \lambda(\tilde{K})\|_1 \leq \sqrt{N} \|\Delta\|_F$. Also, $|\mathrm{Tr}(K) - \mathrm{Tr}(\tilde{K})| = |\mathrm{Tr}(\Delta)| \leq \|\Delta\|_F \|I\|_F = \sqrt{N} \|\Delta\|_F$. Combining and using $\tau \leq \mathrm{Tr}(K), \mathrm{Tr}(\tilde{K})$ yields $\|p - \tilde{p}\|_1 \leq 2\sqrt{N} \|\Delta\|_F / \tau$, hence $(1/2)\|p - \tilde{p}\|_1 \leq \varepsilon$. Fannes–Audenaert then gives $|H(p) - H(\tilde{p})| \leq \varepsilon \log(N-1) + h(\varepsilon)$; exponentiating yields the ratio bound. $\square$

Instantiation for Top- K MoE approximations. Recall $\mathbf{K} = \mathbf{J}\mathbf{J}^\top$ with $\mathbf{J} = [\mathbf{J}^g \mid \mathbf{J}^{\text{exp}}]$ (Section 4.1). The next two corollaries upper bound the perturbation size $\|\Delta\|_F$ in terms of (i) the gate-expert interaction $\mathbf{J}^{g^\top} \mathbf{J}^{\text{exp}}$ and (ii) the gate-only scale $\mathbf{J}^{g^\top} \mathbf{J}^g$.

Corollary F.1.1 (Gate–expert decoupling error bound). *Define the gate–expert decoupled approximation*

$$\mathbf{K}_{\text{bd}} := \mathbf{J}^{\text{g}} \mathbf{J}^{\text{g}\top} + \mathbf{J}^{\text{exp}} \mathbf{J}^{\text{exp}\top}.$$

Then $\Delta_{\text{bd}} := \mathbf{K} - \mathbf{K}_{\text{bd}} = \mathbf{J}^{\text{g}} \mathbf{J}^{\text{exp}\top} + \mathbf{J}^{\text{exp}} \mathbf{J}^{\text{g}\top}$ and

$$\|\Delta_{\text{bd}}\|_F \leq 2\|\mathbf{J}^{\text{g}\top}\mathbf{J}^{\text{exp}}\|_F.$$

Consequently, letting $\tau_{\text{bd}} := \min\{\text{Tr}(\mathbf{K}), \text{Tr}(\mathbf{K}_{\text{bd}})\}$ and $\varepsilon_{\text{bd}} := \sqrt{N}\|\Delta_{\text{bd}}\|_F/\tau_{\text{bd}}$, Lemma F.1 gives a bound on $|\log r_e(\mathbf{K}) - \log r_e(\mathbf{K}_{\text{bd}})|$ whenever $\varepsilon_{\text{bd}} \leq 1 - 1/N$.

Proof. The identity for Δ_{bd} follows by expanding $\mathbf{J}\mathbf{J}^\top$. For the norm bound, use $\|A + A^\top\|_F \leq 2\|A\|_F$ with $A = \mathbf{J}^g \mathbf{J}^{\text{exp}\top}$ and $\|\mathbf{J}^g \mathbf{J}^{\text{exp}\top}\|_F = \|\mathbf{J}^{g\top} \mathbf{J}^{\text{exp}}\|_F$. $\square$

Corollary F.1.2 (Ignoring gating contribution). *Define the expert-only approximation $\mathbf{K}_{\text{exp}} := \mathbf{J}^{\text{exp}} \mathbf{J}^{\text{exp}^\top}$. Then $\Delta_{\text{exp}} := \mathbf{K} - \mathbf{K}_{\text{exp}} = \mathbf{J}^{\text{g}} \mathbf{J}^{\text{g}^\top} + \mathbf{J}^{\text{g}} \mathbf{J}^{\text{exp}^\top} + \mathbf{J}^{\text{exp}} \mathbf{J}^{\text{g}^\top}$ and*

$$\|\Delta_{\text{exp}}\|_F \leq \|\mathbf{J}^g\|_F^2 + 2\|\mathbf{J}^{g^\top} \mathbf{J}^{\text{exp}}\|_F.$$

Consequently, with $\tau_{\text{exp}} := \min\{\text{Tr}(\mathbf{K}), \text{Tr}(\mathbf{K}_{\text{exp}})\}$ and $\varepsilon_{\text{exp}} := \sqrt{N}\|\Delta_{\text{exp}}\|_F/\tau_{\text{exp}}$, Lemma F.1 upper bounds $|\log r_e(\mathbf{K}) - \log r_e(\mathbf{K}_{\text{exp}})|$ whenever $\varepsilon_{\text{exp}} \leq 1 - 1/N$.

Proof. Expand $\mathbf{J}\mathbf{J}^\top$ and apply the triangle inequality. Use $\|\mathbf{J}^g \mathbf{J}^{g^\top}\|_F \leq \|\mathbf{J}^g\|_F^2$ and $\|\mathbf{J}^g \mathbf{J}^{\text{exp}\top} + \mathbf{J}^{\text{exp}} \mathbf{J}^{g^\top}\|_F \leq 2\|\mathbf{J}^g \mathbf{J}^{\text{exp}}\|_F$ as above. $\square$

Corollary F.1.3 (K-FAC factorization error bound). *Let $\mathbf{G}^{\hat{\mathbf{G}}^{\text{N}}, \text{exp}}_{\text{out}} = \frac{1}{T} \sum_t (g_t g_t^\top) \otimes (a_t a_t^\top)$ and $\mathbf{G}^{\hat{\text{exp}}}_{\text{out}} = \frac{1}{T} \sum_t g_t g_t^\top$, $\mathbf{A}^{\hat{\text{exp}}}_{\text{last}} = \frac{1}{T} \sum_t a_t a_t^\top$ (as in Figure 11). Then*

$$\|\mathbf{G}^{\mathbf{GN}, \text{exp}}_{\text{out}} - \mathbf{G}^{\text{exp}}_{\text{out}} \otimes \mathbf{A}^{\text{exp}}_{\text{last}}\|_F \leq \sqrt{\frac{1}{T} \sum_t \|g_t g_t^\top - \mathbf{G}^{\text{exp}}_{\text{out}}\|_F^2} \cdot \sqrt{\frac{1}{T} \sum_t \|a_t a_t^\top - \mathbf{A}^{\text{exp}}_{\text{last}}\|_F^2}.$$

Remark. *K-FAC is accurate when token-wise activation and gradient second-moment factors exhibit weak co-fluctuation across tokens. In particular, the factorization is exact when the centered factors $(g_t g_t^\top - \mathbf{G}^{\text{exp}}_{\text{out}})$ and $(a_t a_t^\top - \mathbf{A}^{\text{exp}}_{\text{last}})$ are uncorrelated in the Hilbert–Schmidt inner product. Moreover, for any perturbation ΔW ,*

$$\left| \langle \Delta W, (\mathbf{G}^{\hat{\mathbf{G}}^{\text{N},\text{exp}}_{\text{out}} - \mathbf{G}^{\hat{\mathbf{G}}^{\text{exp}}_{\text{out}}} \otimes \mathbf{A}^{\hat{\mathbf{A}}^{\text{exp}}_{\text{last}}}) \Delta W \rangle_F \right| \leq \left\| \mathbf{G}^{\hat{\mathbf{G}}^{\text{N},\text{exp}}_{\text{out}}} - \mathbf{G}^{\hat{\mathbf{G}}^{\text{exp}}_{\text{out}}} \otimes \mathbf{A}^{\hat{\mathbf{A}}^{\text{exp}}_{\text{last}}} \right\|_F \|\Delta W\|_F^2,$$

so the same bound controls the deviation between the GN and K-FAC directional curvatures used in Figure 11 (up to normalization by trace).

Proof. Write $\mathbf{G}^{\text{GN}, \text{exp}}_{\text{out}} - \mathbf{G}^{\text{exp}}_{\text{out}} \otimes \mathbf{A}^{\text{exp}}_{\text{last}} = \frac{1}{T} \sum_t (X_t - \bar{X}) \otimes (Y_t - \bar{Y})$ with $X_t = g_t g_t^\top$, $Y_t = a_t a_t^\top$, $\bar{X} = \mathbf{G}^{\text{exp}}_{\text{out}}$, $\bar{Y} = \mathbf{A}^{\text{exp}}_{\text{last}}$. Then $\|(X_t - \bar{X}) \otimes (Y_t - \bar{Y})\|_F = \|X_t - \bar{X}\|_F \|Y_t - \bar{Y}\|_F$ and

$$\left\| \frac{1}{T} \sum_t (X_t - \bar{X}) \otimes (Y_t - \bar{Y}) \right\|_F \leq \frac{1}{T} \sum_t \|X_t - \bar{X}\|_F \|Y_t - \bar{Y}\|_F \leq \sqrt{\frac{1}{T} \sum_t \|X_t - \bar{X}\|_F^2} \sqrt{\frac{1}{T} \sum_t \|Y_t - \bar{Y}\|_F^2},$$

by Cauchy–Schwarz. The quadratic-form bound follows from $|\langle U, MV \rangle_F| \leq \|M\|_F \|U\|_F \|V\|_F$ with $U=V=\Delta W$. $\square$

Connection to the Gauss–Newton blocks. Since $\mathbf{G}^{\text{GN},g,\text{exp}} = (1/N)\mathbf{J}^g{}^\top \mathbf{J}^{\text{exp}}$ and $\mathbf{G}^{\text{GN},g} = (1/N)\mathbf{J}^g{}^\top \mathbf{J}^g$, the perturbation magnitudes in Corollaries F.1.1–F.1.2 are directly controlled by the gate–expert cross-block and the gate-block scale in $\mathbf{G}^{\text{GN}}$. In particular, the gate–expert block cosine used in Figure 8 implies

$$\|\mathbf{G}^{\text{GN},g,\text{exp}}\|_F = \cos(\text{gate}, \text{expert}) \cdot \sqrt{\|\mathbf{G}^{\text{GN},g}\|_F \|\mathbf{G}^{\text{GN},\text{exp}}\|_F},$$

so a small off-diagonal cosine provides a direct quantitative link between the empirical diagnostic and the perturbation terms that govern the induced change in $\log r_e(\mathbf{K})$ via Lemma F.1.

G. Projection onto Multiples of Identity

Lemma G.1 (Frobenius projection onto $\text{span}\{I\}$). *For any $G \in \mathbb{R}^{E \times E}$,*

$$\begin{aligned} \arg \min_{\alpha \in \mathbb{R}} \|G - \alpha I\|_F^2 &= \frac{\text{Tr}(G)}{E}, \\ \min_{\alpha} \|G - \alpha I\|_F^2 &= \|G\|_F^2 - \frac{\text{Tr}(G)^2}{E}. \end{aligned}$$

Proof. Write $\|G - \alpha I\|_F^2 = \text{Tr}(G^\top G) - 2\alpha \text{Tr}(G) + E\alpha^2$, a strictly convex quadratic in α . Setting the derivative to zero gives $\alpha^* = \text{Tr}(G)/E$ and the stated minimum. Equivalently, in the Frobenius inner product $\langle A, B \rangle := \text{Tr}(A^\top B)$, the orthogonal projection of G onto $\text{span}\{I\}$ is $(\langle G, I \rangle / \langle I, I \rangle)I = (\text{Tr}(G)/E)I$. $\square$

H. Additional Experimental Results

This section provides additional experimental results that complement the main paper. We report per-task success rates underlying the main-paper averages, followed by additional ablations, additional robustness analyses, and a brief runtime-overhead summary. Results are reported as mean $\pm$ standard deviation over five seeds.

H.1. Per-Task Success Rates

We report full per-task success rates underlying the average summaries in the main paper, along with additional success-rate comparisons between RL and CRL for alternative baselines. For CRL, we report *per-task final success* evaluated on each task immediately after its own training segment (rather than evaluating the final policy on all tasks after completing the sequence) to eliminate the influence of catastrophic forgetting and isolate plasticity loss.

Table 3. Full per-task final success rates on MetaWorld-10 under RL for dense PPO, a widened dense PPO baseline whose actor MLP has $10\times$ as many parameters (PPO(10x)), and a Top- K MoE actor with different mitigation baselines (SPHERE, C-CHAIN (Tang et al., 2025), continual backpropagation (CBP; Dohare et al., 2024), and a weight-space orthogonality baseline (PW) following Chung et al. (2024)). Figure 4 summarizes the corresponding averages.

Task	PPO	PPO(10x)	Top- K MoE	SPHERE	C-CHAIN	CBP	PW
hammer	0.76 $\pm$ 0.39	1.00 $\pm$ 0.00	0.98 $\pm$ 0.04	0.96 $\pm$ 0.08	0.90 $\pm$ 0.16	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00
push-wall	0.96 $\pm$ 0.05	0.80 $\pm$ 0.28	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	0.78 $\pm$ 0.35	1.00 $\pm$ 0.00	0.90 $\pm$ 0.13
faucet-close	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	0.92 $\pm$ 0.16	1.00 $\pm$ 0.00
push-back	0.18 $\pm$ 0.27	0.33 $\pm$ 0.47	0.00 $\pm$ 0.00	0.02 $\pm$ 0.04	0.20 $\pm$ 0.40	0.16 $\pm$ 0.32	0.00 $\pm$ 0.00
stick-pull	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.16 $\pm$ 0.32	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
handle-press-side	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00
push	0.68 $\pm$ 0.34	0.83 $\pm$ 0.17	0.82 $\pm$ 0.27	1.00 $\pm$ 0.00	0.94 $\pm$ 0.08	0.96 $\pm$ 0.05	0.96 $\pm$ 0.05
shelf-place	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00
window-close	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00
peg-unplug-side	0.92 $\pm$ 0.12	0.65 $\pm$ 0.35	0.90 $\pm$ 0.13	1.00 $\pm$ 0.00	0.84 $\pm$ 0.23	0.84 $\pm$ 0.08	0.86 $\pm$ 0.15
average	0.65 $\pm$ 0.12	0.66 $\pm$ 0.13	0.67 $\pm$ 0.04	0.71 $\pm$ 0.04	0.67 $\pm$ 0.12	0.69 $\pm$ 0.06	0.67 $\pm$ 0.03

Table 4. Full per-task final success rates on MetaWorld-10 under CRL for dense PPO, a widened dense PPO baseline (PPO(10x)), and a Top- K MoE actor with different mitigation baselines (SPHERE, PW, C-CHAIN, and CBP). Figure 4 summarizes the corresponding averages.

Task	PPO	PPO(10x)	Top- K MoE	SPHERE	PW	C-CHAIN	CBP
hammer	1.00 ± 0.00	0.97 ± 0.05	0.98 ± 0.04	0.98 ± 0.04	1.00 ± 0.00	0.80 ± 0.40	0.60 ± 0.45
push-wall	0.20 ± 0.40	0.00 ± 0.00	0.00 ± 0.00	0.70 ± 0.25	0.00 ± 0.00	0.04 ± 0.08	0.18 ± 0.36
faucet-close	0.14 ± 0.23	0.67 ± 0.47	0.38 ± 0.47	1.00 ± 0.00	0.70 ± 0.38	0.38 ± 0.38	1.00 ± 0.00
push-back	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.18 ± 0.36
stick-pull	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
handle-press-side	0.50 ± 0.42	0.45 ± 0.45	0.50 ± 0.45	0.98 ± 0.04	0.62 ± 0.39	0.60 ± 0.45	0.70 ± 0.40
push	0.00 ± 0.00	0.05 ± 0.05	0.00 ± 0.00	0.22 ± 0.17	0.00 ± 0.00	0.00 ± 0.00	0.06 ± 0.08
shelf-place	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00
window-close	0.00 ± 0.00	0.00 ± 0.00	0.20 ± 0.40	1.00 ± 0.00	0.03 ± 0.04	0.40 ± 0.49	0.20 ± 0.40
peg-unplug-side	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.02 ± 0.04	0.00 ± 0.00	0.02 ± 0.04	0.00 ± 0.00
average	0.18 ± 0.11	0.21 ± 0.10	0.21 ± 0.14	0.49 ± 0.06	0.23 ± 0.08	0.22 ± 0.18	0.29 ± 0.21

Table 5. Full per-task final success rates on HumanoidBench under RL with ReLU activations for dense PPO, a widened dense PPO baseline (PPO(10x)), and MoE baselines. Mitigation baselines (SPHERE, PW, C-CHAIN, CBP) are applied to the Top- K MoE actor.

Task	PPO	PPO(10x)	Top- K MoE	Dense-MoE	DS-MoE	SPHERE	PW	C-CHAIN	CBP
stand	0.74 ± 0.20	0.88 ± 0.12	0.84 ± 0.22	0.87 ± 0.05	0.77 ± 0.17	0.95 ± 0.05	0.68 ± 0.15	0.90 ± 0.08	0.83 ± 0.13
walk	0.62 ± 0.12	0.76 ± 0.39	0.46 ± 0.41	0.57 ± 0.40	0.73 ± 0.13	0.75 ± 0.13	0.83 ± 0.15	0.53 ± 0.41	0.50 ± 0.41
pole	0.60 ± 0.09	0.66 ± 0.14	0.58 ± 0.20	0.60 ± 0.40	0.67 ± 0.13	0.83 ± 0.13	0.73 ± 0.12	0.87 ± 0.05	0.63 ± 0.45
slide	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.33 ± 0.17	0.23 ± 0.19	0.00 ± 0.00	0.00 ± 0.00
run	0.76 ± 0.38	0.78 ± 0.15	0.88 ± 0.13	0.85 ± 0.05	0.85 ± 0.15	0.90 ± 0.08	0.65 ± 0.19	0.57 ± 0.40	0.50 ± 0.50
average	0.54 ± 0.16	0.62 ± 0.16	0.55 ± 0.19	0.58 ± 0.18	0.60 ± 0.11	0.75 ± 0.11	0.62 ± 0.16	0.57 ± 0.19	0.49 ± 0.30

Table 6. Full per-task final success rates on HumanoidBench under CRL with ReLU activations for dense PPO, a widened dense PPO baseline (PPO(10x)), and MoE baselines. Mitigation baselines (SPHERE, PW, C-CHAIN, CBP) are applied to the Top- K MoE actor.

Task	PPO	PPO(10x)	Top- K MoE	Dense-MoE	DS-MoE	SPHERE	PW	C-CHAIN	CBP
stand	0.74 ± 0.20	0.88 ± 0.12	0.84 ± 0.22	0.90 ± 0.00	0.77 ± 0.17	0.85 ± 0.18	0.70 ± 0.16	0.90 ± 0.08	0.83 ± 0.13
walk	0.78 ± 0.13	0.74 ± 0.22	0.92 ± 0.08	0.90 ± 0.00	0.83 ± 0.09	0.98 ± 0.04	0.83 ± 0.09	0.73 ± 0.19	0.90 ± 0.00
pole	0.26 ± 0.26	0.42 ± 0.29	0.06 ± 0.08	0.05 ± 0.05	0.33 ± 0.34	0.38 ± 0.18	0.40 ± 0.22	0.23 ± 0.26	0.53 ± 0.41
slide	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.03 ± 0.04	0.07 ± 0.09	0.00 ± 0.00	0.00 ± 0.00
run	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.00 ± 0.00	0.47 ± 0.17	0.00 ± 0.00	0.20 ± 0.20	0.00 ± 0.00
average	0.36 ± 0.12	0.41 ± 0.13	0.36 ± 0.08	0.37 ± 0.01	0.39 ± 0.12	0.54 ± 0.12	0.40 ± 0.11	0.41 ± 0.15	0.45 ± 0.11

H.1.1. A CONTINUAL-LEARNING SUMMARY METRIC ON HUMANOIDBENCH (FIRST FIVE TASKS)

Average success of the final policy. Per-task final success reveals which tasks benefit most, but it can obscure the overall plasticity–retention trade-off in a continual-learning sequence. We therefore report a complementary single-number metric: after sequentially training on the first five HumanoidBench tasks (ending with `h1_run`), for each random seed we take the final checkpoint and evaluate it on *all* five tasks, then average their success rates. This five-task average directly reflects both adaptation to later tasks and retention on earlier tasks in one summary score. Importantly, the improvement does not come at the cost of increased forgetting: SPHERE improves the five-task average while maintaining comparable (and often higher) success on earlier tasks (Table 6).

Table 7. Continual-learning (CL) summary on the first five HumanoidBench tasks. For each random seed, we evaluate the final checkpoint after training on `h1_run` on `h1_stand/walk/pole/slide/run` and report the mean success across these five tasks (mean $\pm$ standard deviation across independent runs).

Method	Avg. (5 tasks)
Top- K MoE	0.03 ± 0.04
SPHERE	0.25 ± 0.05

H.2. Additional Ablation Experiments

H.2.1. SPHERE DESIGN ABLATIONS

Expert-Only Regularization. By design, SPHERE regularizes only the expert representations in the actor and leaves the gate unregularized, to isolate effects on policy plasticity without directly shaping routing. Concretely, it applies the penalty to the routing-weighted expert feature Gram constructed from the weighted concatenation in Eq. (19). We ablate an unweighted variant that drops the routing weights and directly concatenates expert features in Table 8.

Gate-Only Regularization. To isolate whether the gains come from regularizing the expert representations versus directly regularizing routing, we remove the expert penalty and apply the regularizer only to the gate outputs. This improves over the baseline (0.43 ± 0.08 vs. 0.36 ± 0.08) but remains below SPHERE (0.54 ± 0.12), suggesting that expert-feature mode separation is the main driver of the gains.

Block-Diagonal Approximation. Our analysis uses a block-diagonal approximation across layers to obtain a tractable layerwise surrogate objective. As a stress test, we remove this layerwise separation by concatenating the weighted expert features from all expert layers into a single vector and regularizing the resulting Gram. This variant attains 0.47 ± 0.12 , improving over the baseline but providing no additional gain over SPHERE, supporting the adequacy of the block-diagonal layerwise proxy.

Concatenation Across Experts. SPHERE regularizes a Gram matrix formed from the weighted concatenation of features across experts, which couples experts through a shared representation geometry. To test whether this coupling is important, we keep the same layerwise placement as SPHERE but regularize each expert’s features independently (i.e., without concatenating across experts) and sum the losses across experts. This yields 0.40 ± 0.08 , only slightly above the baseline and below SPHERE, suggesting that coupling experts via concatenation is important for the gains.

Layer-Mixture Entropy Regularization. Proposition 4.1 expresses $r_e(\mathbf{K})$ as a mixture over per-layer terms with weights $\{\alpha_\ell^{\text{exp}}\}$. We ablate adding an auxiliary entropy regularizer $-\sum_{\ell: s_\ell^{\text{exp}} > 0} \alpha_\ell^{\text{exp}} \log \alpha_\ell^{\text{exp}}$ on top of SPHERE, which yields 0.41 ± 0.03 . This underperforms SPHERE, suggesting that explicitly shaping the layer-mixture weights is not beneficial in this setting. One possible explanation is that this term imposes a global cross-layer constraint through $\{\alpha_\ell^{\text{exp}}\}$ and can overconstrain optimization, reducing the flexibility of different layers to adapt.

Regularizing Both Feature and Gradient Factors. Our method regularizes the weighted expert feature Gram $\mathbf{A}_{\text{last}}^{\text{exp}}$ (Section 4.2), treating the backprop-gradient Gram $\mathbf{G}_{\text{last}}^{\text{exp}}$ as fixed in the feature–gradient effective-rank factorization (Section 4). As an additional ablation, we apply the same SPHERE penalty to both $\mathbf{A}_{\text{last}}^{\text{exp}}$ and $\mathbf{G}_{\text{last}}^{\text{exp}}$, yielding 0.77 ± 0.23 . This suggests that regularizing gradient factors can further help, although the higher variance indicates less stable optimization. However, $\mathbf{G}_{\text{last}}^{\text{exp}}$ is not directly available from forward passes and requires additional backpropagation to extract, increasing implementation complexity and computational overhead. We therefore do not adopt joint $\mathbf{A}_{\text{last}}^{\text{exp}} - \mathbf{G}_{\text{last}}^{\text{exp}}$ regularization as the main SPHERE instantiation and leave a more systematic study of jointly regularizing $\mathbf{A}_{\text{last}}^{\text{exp}}$ and $\mathbf{G}_{\text{last}}^{\text{exp}}$ to future work.

Regularizing the Kronecker Proxy $\mathbf{A}_{\text{last}}^{\text{exp}} \otimes \mathbf{G}_{\text{last}}^{\text{exp}}$. Section 4 introduces a Kronecker-product proxy $\tilde{\mathbf{G}}_\ell^{\text{GN,exp}} = \mathbf{A}_{\ell-1}^{\text{exp}} \otimes \mathbf{G}_\ell^{\text{exp}}$ for the expert-layer Gauss–Newton blocks. As an additional ablation, we directly apply the SPHERE penalty to the proxy matrix $\mathbf{A}_{\text{last}}^{\text{exp}} \otimes \mathbf{G}_{\text{last}}^{\text{exp}}$, which yields 0.66 ± 0.20 average success. This improves substantially over the baseline and even exceeds SPHERE in mean, suggesting that explicitly shaping a feature–gradient proxy of the expert Gauss–Newton geometry can be beneficial. However, the high variance indicates less stable optimization, consistent with coupling two stochastic factors and requiring extraction of $\mathbf{G}_{\text{last}}^{\text{exp}}$ via additional backpropagation.

Regularizing the Khatri–Rao Expert Block $\mathbf{A}_{\text{last}}^{\text{exp}} * \mathbf{G}_{\text{last}}^{\text{exp}}$. Section 4 models the same-layer expert Gauss–Newton blocks in block Khatri–Rao form $\mathbf{G}_\ell^{\text{GN,exp}} = \mathbf{A}_{\ell-1}^{\text{exp}} * \mathbf{G}_\ell^{\text{exp}}$. As an additional ablation, we directly apply the SPHERE penalty to $\mathbf{A}_{\text{last}}^{\text{exp}} * \mathbf{G}_{\text{last}}^{\text{exp}}$, which yields 0.50 ± 0.25 average success. This variant underperforms SPHERE and is similarly high-variance, suggesting that directly regularizing the full expert-block geometry is either too strict for this setting or introduces additional optimization noise relative to regularizing the feature Gram alone.

Table 8. Additional SPHERE design ablations on HumanoidBench under CRL. All variants use the same setup as SPHERE and differ only in how the penalty is constructed or where it is applied.

Variant	Average success
w/o SPHERE	0.36 ± 0.08
w/ SPHERE	0.54 ± 0.12
Unweighted feature Gram	0.50 ± 0.03
Joint feature Gram across layers	0.47 ± 0.12
Per-expert loss sum (no concatenation)	0.40 ± 0.08
Gate-only regularization	0.43 ± 0.08
Layer-mixture entropy term	0.41 ± 0.03
On $\mathbf{A}_{\text{last}}^{\text{exp}}$ and $\mathbf{G}_{\text{last}}^{\text{exp}}$	0.77 ± 0.23
On $\mathbf{A}_{\text{last}}^{\text{exp}} \otimes \mathbf{G}_{\text{last}}^{\text{exp}}$	0.66 ± 0.20
On $\mathbf{A}_{\text{last}}^{\text{exp}} * \mathbf{G}_{\text{last}}^{\text{exp}}$	0.50 ± 0.25

H.3. Additional Robustness Analysis Experiments

H.3.1. ACTIVATION-FUNCTION ABLATION

Activation Agnosticism. We ablate the policy activation function on HumanoidBench under CRL. Table 9 compares Top- K MoE and SPHERE under each activation, keeping all other settings fixed; the Tanh setting differs from the ReLU baseline only in the activation. SPHERE improves over the baseline under both ReLU and Tanh, suggesting that the gains do not rely on a particular activation choice.

Table 9. Activation-function ablation on HumanoidBench under CRL. We report average success on the five-task suite; SPHERE consistently improves over Top- K MoE under both ReLU and Tanh, indicating robustness to the activation choice.

Activation	Top- K MoE	SPHERE
ReLU	0.36 ± 0.08	0.54 ± 0.12
Tanh	0.36 ± 0.10	0.58 ± 0.17

H.3.2. STRUCTURE AGNOSTICISM

Structure Agnosticism. To test whether SPHERE depends on a particular MoE structure, we apply the same penalty to Dense-MoE and DS-MoE actors under CRL. Results are summarized in Table 10. SPHERE attains 0.55 ± 0.12 on Dense-MoE and 0.51 ± 0.08 on DS-MoE, suggesting that the method transfers across MoE variants beyond Top- K .

Table 10. Structure-agnosticity on HumanoidBench under CRL. Values are average success (mean $\pm$ standard deviation) over five seeds; applying SPHERE improves both Dense-MoE and DS-MoE, indicating the penalty transfers beyond Top- K MoE.

MoE variant	w/o SPHERE	w/ SPHERE
Dense-MoE	0.37 ± 0.01	0.55 ± 0.12
DS-MoE	0.39 ± 0.12	0.51 ± 0.08

H.3.3. PENALTY RATIO ROBUSTNESS

Penalty-ratio robustness. SPHERE sets the adaptive regularization coefficient as $\lambda^e = \rho \cdot \text{stopgrad}\left(\frac{\|\nabla_{\theta} \mathcal{L}_{\text{actor}}\|_2}{\|\nabla_{\theta} \mathcal{L}_{\text{SPHERE}}\|_2 + \epsilon}\right)$. Figure 7 reports a sweep over ρ on HumanoidBench under CRL. Performance is best at $\rho = 1\text{e-}3$, while $\rho = 0$ removes the regularizer and recovers the baseline. Importantly, all nonzero ρ values in this sweep outperform $\rho = 0$ by a clear margin ($+0.09$ to $+0.17$ average success), indicating that SPHERE is robust to the choice of ρ and provides consistent gains once the regularizer is enabled. Overall, performance exhibits a broad optimum around 10^{-3} . When ρ is increased (e.g., $5\text{e-}2$), performance degrades relative to the best setting, consistent with over-regularization that can overconstrain representation learning. We additionally find that this robustness carries over across benchmarks: on MetaWorld under CRL, using the same $\rho=1\text{e-}3$ as HumanoidBench (two orders of magnitude smaller than our default MetaWorld setting in Table 2) yields 0.369 ± 0.130 average success, still improving over the unregularized Top- K MoE baseline (0.21 ± 0.14 ; Table 4).

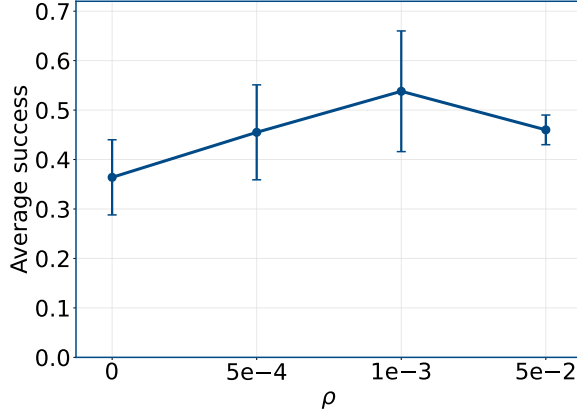


Figure 7. Effect of the SPHERE ratio hyperparameter ρ on HumanoidBench under CRL. Points show mean $\pm$ standard deviation of average success over five seeds; all nonzero ρ values in the sweep outperform $\rho = 0$ (by +0.09 to +0.17 average success), indicating robustness, with a broad optimum around $\rho = 1e-3$.

H.3.4. ALGORITHM AGNOSTICISM

Algorithm Agnosticism. To test whether SPHERE is algorithm-agnostic, we apply the same penalty to a Top- K MoE critic trained with TD3 under CRL. Table 11 reports mean and standard deviation of final success over seeds. SPHERE improves average success from 0.50 ± 0.08 to 0.65 ± 0.09 , suggesting that the penalty can also benefit off-policy training. This further suggests that SPHERE is not limited to MoE actors and can also benefit MoE critics.

Table 11. Algorithm-agnosticity on MetaWorld under CRL using TD3. Values are average final success (mean $\pm$ standard deviation) over seeds; SPHERE improves success for a Top- K MoE critic, indicating it can benefit off-policy training and is not limited to PPO actors.

Method	Average success
Top- K MoE critic (TD3)	0.50 ± 0.08
Regularized MoE critic (TD3)	0.65 ± 0.09

H.4. SPHERE Promotes Expert Specialization

We study whether SPHERE increases expert specialization on HumanoidBench by measuring (i) representation-level diversity among the selected experts and (ii) geometric coherence of the gate’s routing in its embedding space.

Representation-level specialization. Let $h_{i,e} \in \mathbb{R}^H$ denote the last-hidden feature of expert e at state s_i , and let S_i denote the set of selected experts at s_i . We measure redundancy among selected experts via the mean cosine overlap over pairs $(e, e') \in S_i$ with $e \neq e'$. We also report a complementary orthogonality score defined as the mean of $1 - |\cos|$ over the same pairs.

Gate geometry and local consistency. Let g_i be the gate embedding at s_i and let y_i be the Top-1 expert label. We compute (i) the silhouette score of $\{g_i\}$ clustered by $\{y_i\}$, and (ii) a k_{nn} -nearest-neighbor consistency score that averages the Jaccard overlap between the selected-expert sets of neighboring states in the gate-embedding space.

Expert load balancing. Let c_e denote the number of times expert e is selected across the batch, i.e., $c_e := |\{(i, e') : e' \in S_i, e'=e\}|$, so that $\sum_e c_e = \sum_i |S_i|$. We quantify load imbalance by the maximum relative deviation from the uniform expected count,

$$\text{MaxVio} := \max_e \frac{|c_e - \bar{c}|}{\bar{c}}, \quad \bar{c} := \frac{1}{E} \sum_{e=1}^E c_e,$$

We also report the active-expert ratio $|\{e : c_e > 0\}|/E$ as a simple coverage measure, and the normalized routing entropy

$$H_{\text{route}} := \frac{-\sum_{e=1}^E \hat{p}_e \log \hat{p}_e}{\log E}, \quad \hat{p}_e := \frac{c_e}{\sum_{e'=1}^E c_{e'}}.$$

Results. Table 12 reports these metrics on HumanoidBench `h1_run` using $N=512$ deterministic on-policy rollout states across independent runs. SPHERE reduces overlap and increases orthogonality, indicating less redundant expert representations for the same state. In parallel, SPHERE improves gate-embedding coherence and local routing consistency. Overall, these results indicate that SPHERE improves representation-level specialization and routing geometry in MoE actors.

Table 12. Expert specialization and load-balancing indicators on HumanoidBench `h1_run`. Values are mean $\pm$ standard deviation across independent runs on $N=512$ deterministic on-policy rollout states. Lower overlap and higher orthogonality indicate stronger representation-level specialization among selected experts. Higher silhouette and higher kNN Jaccard overlap indicate more coherent gate geometry and locally consistent routing. Lower MaxVio, higher active-expert ratio, and higher routing entropy indicate more uniform expert usage.

Method	Overlap $\downarrow$	Orthogonality $\uparrow$	Silhouette $\uparrow$	kNN Jaccard $\uparrow$	MaxVio $\downarrow$	Active ratio $\uparrow$	Routing entropy $\uparrow$
Top- K MoE (baseline)	0.34 ± 0.01	0.66 ± 0.01	0.02 ± 0.15	0.82 ± 0.11	2.73 ± 0.85	0.67 ± 0.21	0.66 ± 0.11
SPHERE (ours)	0.13 ± 0.01	0.87 ± 0.01	0.15 ± 0.06	0.96 ± 0.02	2.70 ± 0.75	0.69 ± 0.10	0.51 ± 0.08

H.5. Runtime Overhead

Table 13 reports wall-clock training time per task as a function of the expert count. Across $E \in \{10, 20, 40\}$, the Top- K MoE baseline remains essentially flat, and SPHERE adds only a 4.9–6.3% overhead, indicating that the SPHERE regularizer is not a performance bottleneck.

Efficient Gram computation. Recall that the SPHERE penalty depends only on $\text{Tr}(\mathbf{A}_{\text{last}}^{\text{exp}})$ and $\|\mathbf{A}_{\text{last}}^{\text{exp}}\|_F^2$. We therefore evaluate it using whichever is cheaper to form: $\mathbf{A}_{\text{last}}^{\text{exp}}$ or the sample Gram $(1/N)\Phi\Phi^\top$. By cyclicity of trace and the fact that $\Phi^\top\Phi$ and $\Phi\Phi^\top$ share the same non-zero eigenvalues, we have $\text{Tr}(\mathbf{A}_{\text{last}}^{\text{exp}}) = \text{Tr}((1/N)\Phi\Phi^\top)$ and $\|\mathbf{A}_{\text{last}}^{\text{exp}}\|_F^2 = \|(1/N)\Phi\Phi^\top\|_F^2$, yielding cost $\mathcal{O}(\min\{N, m\}^2 \max\{N, m\})$, i.e., $\mathcal{O}(N^2m)$ when $m \gg N$. Since m scales linearly with the number of experts for fixed per-expert width, the Gram computation scales linearly in the expert count.

Block-sparse acceleration. In our regularizer, gating plays an additional role beyond selecting experts: because the penalty is defined on the *gating-weighted* expert features, Top- K routing induces an explicit block-sparse structure in Φ (and hence in the expert-layer Gram), which can be exploited to accelerate the computation. Concretely, Top- K routing makes each row of Φ block-sparse: writing $\Phi = [\Phi_1, \dots, \Phi_E]$ with $\Phi_e \in \mathbb{R}^{N \times d}$ (so $m = Ed$), sample i activates only a set $S_i \subseteq [E]$ with $|S_i| = K$, and its feature row decomposes as $\phi_i = [\phi_{i,1}; \dots; \phi_{i,E}]$ where $\phi_{i,e} = 0$ if $e \notin S_i$. This yields

$$\text{Tr}((1/N)\Phi\Phi^\top) = \frac{1}{N} \sum_{i=1}^N \|\phi_i\|_2^2 = \frac{1}{N} \sum_{i=1}^N \sum_{e \in S_i} \|\phi_{i,e}\|_2^2,$$

and the sample Gram entries admit a co-activation form

$$\left(\frac{1}{N}\Phi\Phi^\top\right)_{ij} = \frac{1}{N} \sum_{e \in S_i \cap S_j} \langle \phi_{i,e}, \phi_{j,e} \rangle.$$

Consequently, both $\text{Tr}(\mathbf{A}_{\text{last}}^{\text{exp}})$ and $\|\mathbf{A}_{\text{last}}^{\text{exp}}\|_F^2 = \|(1/N)\Phi\Phi^\top\|_F^2 = \sum_{i,j} ((1/N)\Phi\Phi^\top)_{ij}^2$ can be evaluated without materializing dense Φ by accumulating only over co-activated sample pairs. In particular, if $\mathcal{I}_e := \{i : e \in S_i\}$, we can compute all nonzero Gram entries by iterating experts and accumulating $\langle \phi_{i,e}, \phi_{j,e} \rangle$ for $(i, j) \in \mathcal{I}_e^2$, which costs $\mathcal{O}\left(\sum_{e=1}^E |\mathcal{I}_e|^2 d\right)$ time and uses $\mathcal{O}(Nkd)$ storage for the active blocks plus an $\mathcal{O}(N^2)$ accumulator for the sample Gram. Under approximately balanced routing ($|\mathcal{I}_e| \approx NK/E$), this gives an expected cost $\mathcal{O}(N^2K^2d/E) = \mathcal{O}(N^2K^2m/E^2)$, which can be substantially smaller than the dense $\mathcal{O}(N^2m)$ baseline when $K \ll E$.

Table 13. Wall-clock training time measured on NVIDIA A100. Values are averaged over the five-task suite with 10M environment steps per task, reported as hours per task for different expert counts.

Method	$E=10$	$E=20$	$E=40$
Top- K MoE (baseline)	2.23	2.23	2.24
SPHERE (ours)	2.37 (+6.28%)	2.34 (+4.93%)	2.37 (+5.80%)

H.6. Validation of Theoretical Approximation Assumptions

Our analysis adopts a block-diagonal approximation of the Gauss–Newton matrix that separates gate and expert parameters (Eq. (14)). For standard dense MLPs, block-diagonal and within-layer independence approximations are widely adopted in practical curvature modeling and natural-gradient methods (e.g., KFAC and its variants; Martens & Grosse, 2015; Grosse & Martens, 2016; Botev et al., 2017). In MoE actors, however, the validity of an analogous *gate–expert* block structure is *not* obvious: Top- K routing multiplies gate probabilities with expert activations, which can in principle induce substantial cross-block curvature between gate and expert parameters. Complementing the empirical diagnostics below, Appendix F provides a general perturbation bound that translates small gate–expert coupling into a controlled change in $r_e(\mathbf{K})$.

Validation of the gate–expert block-diagonal assumption. To empirically assess this MoE-specific assumption, we measure the relative magnitude of the gate–expert cross-block in the Gauss–Newton matrix. Let G_{ab}^{GN} denote the Gauss–Newton sub-block between parameter groups a and b (with $a, b \in \{\text{gate}, \text{expert}\}$). We report the normalized block “cosine similarity”

$$\cos(a, b) := \frac{\|G_{ab}^{\text{GN}}\|_F}{\sqrt{\|G_{aa}^{\text{GN}}\|_F \|G_{bb}^{\text{GN}}\|_F}} \in [0, 1],$$

which is scale-invariant and directly compares cross-block curvature to the geometric mean of the corresponding within-block curvature. If gate and expert parameters were strongly coupled in G^{GN} , we would expect a large off-diagonal value (comparable to the diagonal blocks); conversely, values near zero indicate a near block-diagonal structure. Figure 8 shows that the off-diagonal gate–expert coupling is substantially smaller than the within-block terms in a representative late-stage checkpoint (HumanoidBench h1_run), supporting the gate–expert block-diagonal surrogate used throughout our analysis.

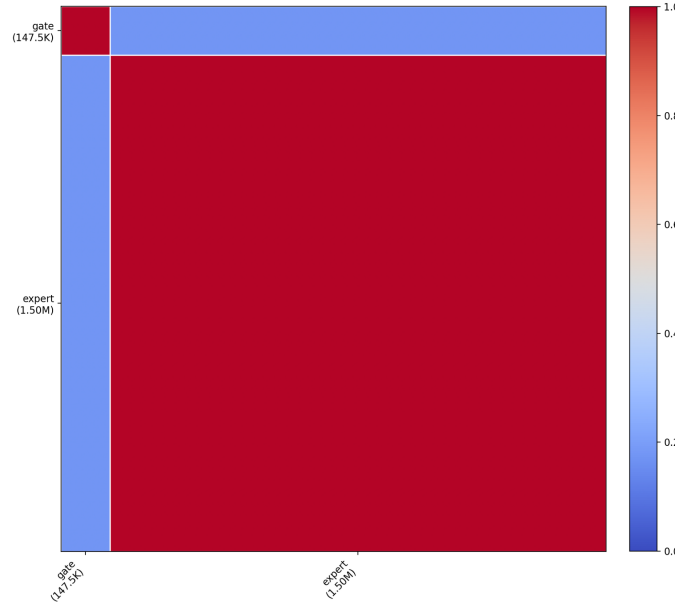


Figure 8. Validation of the gate–expert block-diagonal approximation in a Top- K MoE actor. We visualize the Gauss–Newton block cosine similarity $\cos(a, b)$ between the gate and expert parameter groups on HumanoidBench h1_run. Blocks are ordered as $\{\text{gate}, \text{expert}\}$ to match Eq. (14), and block widths/heights are proportional to parameter counts. A small off-diagonal value indicates weak gate–expert coupling relative to within-block curvature, supporting the use of a gate–expert block-diagonal surrogate in the analysis of spectral plasticity.

Expert-feature isotropy as a proxy for spectral plasticity. Our method regularizes the expert feature Gram $\mathbf{A}_{\text{last}}^{\text{exp}}$ to counteract low-rank simplicity bias in the expert representation. This design is motivated by our analysis, which links spectral plasticity $r_e(\mathbf{K})$ to expert-layer curvature via G^{GN} and further decomposes the expert blocks into feature and gradient factors. Empirically, it is still useful to verify that changes in the feature factor tracked by $r_e(\mathbf{A}_{\text{last}}^{\text{exp}})$ co-vary with the sample-space eNTK effective rank $r_e(\mathbf{K})$ on the same policy.

To this end, we fix a trained MoE actor checkpoint on HumanoidBench h1_run, roll out the policy to collect a large pool of

states, and repeatedly sample batches of $N=64$ states. For each batch, we compute the feature-Gram effective rank $r_e(\mathbf{A}_{\text{last}}^{\text{exp}})$ from the gating-weighted expert features and the eNTK effective rank $r_e(\mathbf{K})$ from the actor’s mean-action Jacobian. Figure 9 shows a clear positive association between the two quantities across batches in a representative checkpoint, suggesting that increasing the isotropy/effective rank of the expert feature Gram provides a meaningful proxy signal for improving spectral plasticity.

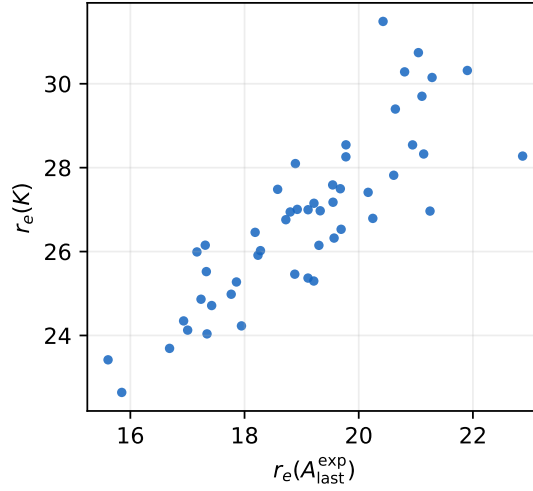


Figure 9. Batchwise relationship between the expert feature-Gram effective rank and the eNTK effective rank in a Top- K MoE actor. We roll out a fixed HumanoidBench `h1_run` policy, sample many batches of $N=64$ states, and compute one point per batch: $x=r_e(\mathbf{A}_{\text{last}}^{\text{exp}})$ and $y=r_e(\mathbf{K})$. The strong positive correlation supports using the SPHERE penalty on $\mathbf{A}_{\text{last}}^{\text{exp}}$ as a practical surrogate for improving spectral plasticity.

Validation of Proposition 4.3. Proposition 4.3 replaces the intractable expert-layer block $\mathbf{G}_{\ell}^{\text{GN},\text{exp}} = \mathbf{A}_{\ell}^{\text{exp}} * \mathbf{G}_{\ell}^{\text{exp}}$ (block Khatri–Rao form) with its Kronecker proxy $\tilde{\mathbf{G}}_{\ell}^{\text{GN},\text{exp}} = \mathbf{A}_{\ell}^{\text{exp}} \otimes \mathbf{G}_{\ell}^{\text{exp}}$, and predicts that optimizing the proxy effective rank is sufficient for improving the expert-layer effective rank under the model. We empirically verify this prediction by checking that the proxy effective rank tracks the target effective rank across rollout batches.

Concretely, we roll out a fixed HumanoidBench `h1_run` checkpoint, repeatedly sample batches of $N=64$ states, and instantiate Proposition 4.3 at the actor’s output block. For each batch, we compute the proxy effective rank $r_e(\tilde{\mathbf{G}}_{\text{out}}^{\text{GN},\text{exp}})$ and the target effective rank $r_e(\mathbf{G}_{\text{out}}^{\text{GN},\text{exp}})$, where $\tilde{\mathbf{G}}_{\text{out}}^{\text{GN},\text{exp}} = \mathbf{A}_{\text{last}}^{\text{exp}} \otimes \mathbf{G}_{\text{out}}^{\text{exp}}$ and $\mathbf{G}_{\text{out}}^{\text{GN},\text{exp}} = \mathbf{A}_{\text{last}}^{\text{exp}} * \mathbf{G}_{\text{out}}^{\text{exp}}$ under the model. Figure 10 shows a strong positive association, supporting the Kronecker proxy as a faithful surrogate for analyzing and optimizing expert-layer spectral properties.

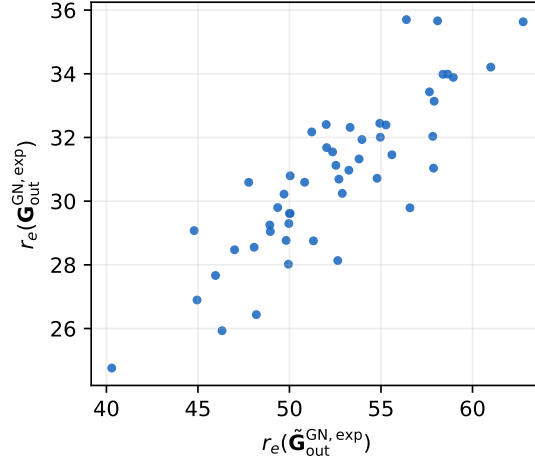


Figure 10. Validation of Proposition 4.3 on a Top- K MoE actor. Each point corresponds to one batch of $N=64$ rollout states from a fixed HumanoidBench h1_run checkpoint. The horizontal axis is the proxy effective rank $r_e(\hat{\mathbf{G}}_{\text{out}}^{\text{GN}, \text{exp}})$, and the vertical axis is the target effective rank $r_e(\mathbf{G}_{\text{out}}^{\text{GN}, \text{exp}})$. The strong monotonic relationship supports the Kronecker proxy as a faithful surrogate at the expert-layer block.

Validation of the K-FAC factorization assumption. Beyond the Kronecker proxy at the level of effective ranks, K-FAC additionally assumes that, within an expert layer, the activation and backprop factors are weakly coupled across tokens (tokens are (state, expert) pairs), so that the empirical GN block $\mathbf{G}_{\text{out}}^{\text{GN}, \text{exp}} = \frac{1}{T} \sum_t (g_t g_t^\top) \otimes (a_t a_t^\top)$ is well-approximated by $\hat{\mathbf{G}}_{\text{out}}^{\text{exp}} \otimes \hat{\mathbf{A}}_{\text{last}}^{\text{exp}}$. To test this without forming either matrix, we sample many random rank-one parameter perturbations ΔW over rollout batches and compare the resulting normalized directional curvatures. Figure 11 compares the K-FAC proxy curvature q_{KFAC} against the empirical GN curvature q_{GN} ; if the factorization holds, the points concentrate along the diagonal.

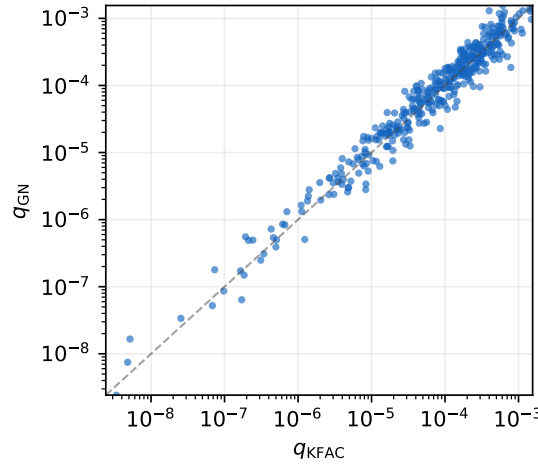


Figure 11. K-FAC factorization validation on the expert output block. Each point corresponds to one rollout batch ($N=64$ states) and one random rank-one perturbation ΔW of the expert output-layer kernel. We evaluate two normalized quadratic forms: $q_{\text{KFAC}} = \langle \Delta W, (\hat{\mathbf{G}}_{\text{out}}^{\text{exp}} \otimes \hat{\mathbf{A}}_{\text{last}}^{\text{exp}}) \Delta W \rangle_F / \text{Tr}(\hat{\mathbf{G}}_{\text{out}}^{\text{exp}} \otimes \hat{\mathbf{A}}_{\text{last}}^{\text{exp}})$ and $q_{\text{GN}} = \langle \Delta W, \mathbf{G}_{\text{out}}^{\text{GN}, \text{exp}} \Delta W \rangle_F / \text{Tr}(\mathbf{G}_{\text{out}}^{\text{GN}, \text{exp}})$. Concentration near the diagonal indicates that the K-FAC proxy accurately predicts the empirical GN curvature in our MoE setting.